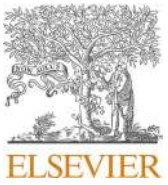
Mechanical Systems and Signal Processing 258 (2026) 114642



Contents lists available at ScienceDirect

Mechanical Systems and Signal Processing

journal homepage: www.elsevier.com/locate/ymssp



Full Length Article



# A hybrid deep learning framework for reliable and interpretable remaining life estimation in turbofan Engines: Focusing on Robustness, Generalization, and Sensor-Level interpretability

Yıldırım Özüpak *

*Department of Electricity and Energy, Silvan Vocational School, Dicle University, 21000 Diyarbakır, Turkey*

ARTICLE INFO

*Keywords:*
Turbofan engine
Hybrid CNN–GRU
Deep learning
Reliability analysis
C-MAPSS

ABSTRACT

Reliable prediction of Remaining Useful Life (RUL) is crucial for improving maintenance planning and operational safety in turbofan engines. This study proposes a hybrid Convolutional Neural Network–Gated Recurrent Unit (CNN–GRU) framework for prognostic analysis using the NASA C-MAPSS benchmark dataset. The CNN layers are employed to extract local degradation-related features from multivariate sensor measurements, while the GRU structure captures temporal degradation trends over long operational sequences. Beyond conventional accuracy-focused evaluations, the proposed framework emphasizes robustness, reproducibility, and interpretability through a reliability-oriented experimental strategy. Comparative analyses were conducted against standalone CNN and GRU models together with component-level ablation experiments and repeated training runs to assess statistical stability. In addition, SHAP and LIME methods were utilized to provide global and local explanations of model behavior and sensor importance. Experimental validation across FD001–FD004 subsets demonstrates that the hybrid CNN–GRU model achieves competitive prediction accuracy with improved consistency under varying operating conditions and fault scenarios. The explainability analysis further reveals strong dependence on degradation-sensitive thermodynamic sensor signals. Overall, the proposed framework offers an interpretable and reproducible deep learning approach for benchmark aero-engine RUL prediction while also acknowledging limitations related to real-world industrial deployment.

## 1. Introduction

In contemporary industrial systems, unplanned equipment failure poses serious safety concerns and financial risks, particularly in safety–critical fields such as aeronautical engineering. Unexpected failures in these settings can result in severe safety repercussions, substantial maintenance costs, and operating delays. Therefore, PHM systems, which track system health, identify degradation early on, and anticipate problems before they happen, have attracted increasing attention. A major goal of PHM is accurately estimating RUL (the number of operating cycles a system can run before failing) [1].

Because degradation processes are complicated and stochastic, RUL prediction is intrinsically difficult. In real-world scenarios, systems are monitored using noisy and high-dimensional sensor data, function under a variety of situations, and show varied degrees of

* Corresponding author.
*E-mail address:* yildirim.ozupak@dicle.edu.tr.

https://doi.org/10.1016/j.ymssp.2026.114642
Received 20 April 2026; Received in revised form 27 May 2026; Accepted 28 June 2026
Available online 8 July 2026
0888-3270/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

initial wear. RUL estimation is a challenging learning issue because of these characteristics, which produce highly nonlinear and nonstationary time-series data [2]. Furthermore, models that can concurrently capture different temporal scales are necessary because deterioration patterns frequently incorporate both short-term local fluctuations and long-term temporal relationships [3].

Although hybrid CNN–GRU architectures have been investigated in the past, their assessment has mostly been restricted to comparisons that are accuracy-focused [4]. The integration of explainability, reproducibility, and reliability analysis into a single framework is what makes this study new, not the architecture itself. In particular, this study fills important gaps in the field by introducing: (i) statistical robustness evaluation over several independent runs; (ii) coupled SHAP and LIME explainability analysis; and (iii) engineering-oriented interpretation of learnt features.

Early methods for predicting RUL depended on physics-based models or conventional machine learning methods like random forests and support vector machines [5]. Although these techniques can work well in controlled environments, they usually need a lot of feature engineering and have trouble generalizing to other operating environments [6]. Because deep learning techniques can automatically learn representations from unprocessed sensor data, they have become more potent alternatives in recent years [7]. While Recurrent Neural Networks (RNNs), such as LSTM and GRU networks, have shown a good ability to simulate long-term temporal dependencies, CNNs have been routinely employed to extract local degradation features [8].

Even with these developments, current methods still have significant drawbacks. While recurrent models may be susceptible to sensor noise and lack explicit procedures for local feature abstraction, CNN-based models frequently fall short in capturing long-range temporal correlations [9]. Hybrid architectures that combine convolutional and recurrent components have been the subject of recent research; however, many of these studies primarily concentrate on predictive accuracy, paying little attention to robustness across several training runs, systematic qualitative analysis, or model explainability [10]. Such omissions can impede practical adoption in safety–critical applications since decision-makers need dependable and understandable models in addition to accurate predictions [11].

Although the RUL literature has previously investigated hybrid CNN–GRU architectures, their evaluation is frequently restricted to single-run accuracy comparisons on the simplest benchmark subset. On the other hand, rather than asserting a fundamentally novel network design, the current study concentrates on methodological rigor. Therefore, the primary contribution is to create a more dependable experimental framework for hybrid CNN–GRU-based RUL prediction using engineering-oriented explainability, controlled benchmarking, corrected ablation logic, robustness analysis across repeated runs, and cross-subset evaluation on FD001–FD004.

The primary novelty of the present study does not lie in proposing a fundamentally new neural architecture, but rather in establishing a rigorously evaluated reliability-oriented prognostic framework for hybrid deep learning-based RUL prediction. Unlike many existing CNN–GRU studies that primarily emphasize single-run predictive accuracy, the proposed framework integrates repeated-run statistical robustness analysis, explainability-driven interpretation, controlled ablation methodology, cross-subset validation, and engineering-oriented degradation interpretation within a unified experimental pipeline.

Accordingly, the main contributions of the present study can be summarized as follows:

- A standardized and reproducible experimental framework is developed to enable consistent comparison among CNN, GRU, and hybrid CNN–GRU architectures under identical data preprocessing, training configuration, optimization strategy, and evaluation criteria;
- The proposed prognostic framework is systematically validated across all four NASA C-MAPSS subsets (FD001–FD004), allowing comprehensive assessment of model generalization capability under diverse operating conditions and fault scenarios;
- A component-oriented ablation strategy is designed in which architectural modifications are performed under controlled and configuration-consistent conditions, thereby reducing potential bias arising from inconsistent model settings;
- Prediction robustness and training stability are investigated through repeated experimental runs, and the results are statistically reported using mean values and standard deviations instead of relying exclusively on single-run performance indicators;

To further clarify the methodological contribution beyond architectural design, Table 1 summarizes the main differences between the present study and representative CNN–GRU-based RUL prediction studies. Unlike many previous works that primarily focus on single-run predictive accuracy on FD001, the proposed framework integrates robustness analysis, explainability evaluation, controlled benchmarking, and engineering-oriented interpretation within a unified reliability-oriented evaluation pipeline.

As shown in Table 1, the proposed study emphasizes methodological reliability, interpretability, and robustness-oriented evaluation rather than introducing a fundamentally new neural architecture.

The remainder of this paper is organized as follows. Section 2 reviews related work on RUL prediction and deep learning-based

**Table 1**
Methodological positioning of the proposed framework against typical CNN–GRU-based RUL prediction studies.

| Aspect | Typical CNN–GRU Studies | Proposed Study |
|---|---|---|
| Single-run evaluation | ✓ | × |
| Multi-run robustness analysis | Limited | ✓ |
| Explainability analysis | Partial | SHAP + LIME |
| Cross-subset validation | Often FD001 only | FD001–FD004 |
| Engineering interpretation | Limited | ✓ |
| Reliability-oriented evaluation | Rare | ✓ |

prognostic methods. Section 3 describes the dataset, proposed models, and experimental methodology. Section 4 presents the experimental results and analysis. Section 5 discusses the findings and their implications, and Section 6 concludes the paper with directions for future research.

## 2. Literature Review

Predicting RUL accurately has grown in importance in prognostic research, especially for turbofan engines running in safety–critical scenarios. Previous research mostly used traditional, physics-based machine learning techniques, which frequently have trouble capturing intricate nonlinear degradation behavior [12]. The potential of deep learning techniques to automatically extract degradation-related characteristics from raw time-series signals has drawn more attention as multivariate sensor data becomes more widely available [7]. As a result, convolutional, recurrent, and hybrid architectures for RUL prediction have been thoroughly studied in recent research, with notable gains in predictive performance.

Recent studies have extensively investigated deep learning-based approaches for turbofan engine RUL prediction using the NASA C-MAPSS dataset. Kovilpillai et al. evaluated CNN, LSTM, GRU, and autoencoder architectures for prognostic analysis and demonstrated the effectiveness of deep learning methods for modeling degradation behavior [13]. Hossain et al. employed CNNs directly on raw sensor sequences without manual feature engineering and achieved improved predictive accuracy compared with conventional approaches [14]. Asif et al. proposed an enhanced preprocessing strategy combined with deep LSTM networks and achieved competitive performance across all C-MAPSS subsets through improved degradation modeling and sensor selection [15]. Sharma et al. highlighted the importance of recurrent architectures and attention-enhanced hybrid frameworks for capturing temporal degradation dependencies [16]. Similarly, Thakkar and Chaoui introduced a DL-RNN structure that outperformed several conventional neural network models on benchmark datasets [17], while Peringal et al. showed that LSTM architectures consistently surpassed MLP models in learning sequential degradation behavior [18].

In addition to deep recurrent models, several studies explored alternative prognostic strategies. Gupta et al. optimized Random Forest models using feature engineering techniques [19], whereas Freitas and Guimarães proposed a computationally efficient fuzzy time-series framework using symbolic sensor representations [20]. Shen et al. improved optimization stability through a modified Gray Wolf Optimizer integrated with deep learning models [21]. Transformer-based architectures have also gained attention due to their strong temporal representation capability. Wu et al. proposed an improved Transformer framework integrated with a Digital Twin environment for real-time prognostic decision support [22]. More recent research trends emphasize interpretability, adaptability, and deployment-oriented prognostic systems [23]. Ensemble deep learning models [24] adaptive clustering strategies, autoencoder-based unsupervised frameworks [25], Gaussian-mixture methods [26], and dilated temporal architectures have all demonstrated promising performance for handling complex degradation trajectories and varying operating conditions [27]. These developments indicate that current research increasingly prioritizes robustness [24, explainability [28], adaptability [29], and engineering usability in addition to predictive accuracy [30].

Recent prognostic studies have increasingly focused on robustness-oriented and domain-generalized RUL prediction frameworks rather than solely maximizing predictive accuracy [31]. For example, dense convolutional regression architectures combined with data augmentation strategies have demonstrated improved robustness against degradation-pattern variability and limited training diversity [31]. Similarly, federated and domain-generalization-based approaches such as CEC-FedISDG have emphasized invariance learning and cross-domain adaptation capability for industrial prognostic systems operating under heterogeneous conditions [32]. These studies indicate that modern RUL research increasingly prioritizes robustness, transferability, uncertainty resistance, and domain adaptability in addition to conventional prediction performance [33]. Motivated by these developments, the present study incorporates repeated-run robustness evaluation, explainability analysis, and cross-subset validation within a unified REE-based prognostic framework.

Using the NASA C-MAPSS dataset, hybrid CNN–GRU architectures for RUL prediction have been thoroughly investigated in recent research. For example, a number of studies have shown that the performance of temporal modeling and feature extraction is enhanced when convolutional layers and recurrent units are combined. Predictive accuracy is the main focus of these investigations, which usually use FD001 or several subsets (FD001–FD004) and RMSE-based evaluation. Nevertheless, the majority of these methods do not include thorough explainability frameworks and do not systematically assess model resilience over several runs. Moreover, the interpretation of learned features at the technical level is frequently neglected. The current study, on the other hand, sets itself apart by combining dual explainability (SHAP + LIME), unified benchmarking, and robustness analysis into a single framework.

## 3. Material and method

The dataset, model architectures, and experimental techniques used for turbofan engine RUL prediction are described in this section. Three deep learning models a CNN, a GRU, and a hybrid CNN–GRU architecture are examined using multivariate time-series sensor data from the NASA C-MAPSS benchmark dataset. Data preprocessing, RUL labeling, model training, and performance assessment using conventional regression measures are all part of the suggested methodology. For safety–critical prognostic applications, explainability approaches are used in addition to forecast accuracy to investigate model behavior and improve transparency.

### 3.1. Dataset description

PHM is essential to contemporary industrial systems because it allows for accurate prediction of the remaining useful life of

complex technical assets and early fault identification. The NASA C-MAPSS dataset is one of the most popular publicly accessible prognostic benchmarks for RUL prediction algorithm development and evaluation [34]. The NASA C-MAPSS dataset was generated using a high-fidelity turbofan engine simulator developed at NASA Ames Research Center to represent realistic degradation processes under different operating conditions. The dataset contains multivariate time-series sensor measurements collected from multiple engines operating until failure. In the training subsets, complete degradation trajectories are available, whereas the test subsets terminate before failure, requiring prediction of the remaining operational cycles after the final observation. To evaluate prognostic models under varying levels of complexity, the dataset is divided into four subsets: FD001, FD002, FD003, and FD004. These subsets differ according to the number of operating conditions and fault modes. FD001 represents the simplest case with a single operating condition and fault type, while FD004 contains multiple operating conditions and concurrent fault mechanisms, making it the most challenging subset. The main characteristics of the dataset subsets are summarized in Table 2.

A two-spool turbofan engine architecture with a fan, low-pressure compressor (LPC), high-pressure compressor (HPC), combustor, high-pressure turbine (HPT), and low-pressure turbine (LPT) is represented by the simulated turbofan engines in the C-MAPSS dataset. The High-Pressure Compressor (HPC), a crucial element affecting engine performance and efficiency, is the main target of the degradation modeled in FD001. While other subsets contain different operating regimes including altitude, Mach number, and temperature conditions, the engines in FD001 run under steady-state settings. Sensor readings like temperature, pressure, and rotational speed are directly impacted by the dataset's simulation of actual degradation mechanisms like efficiency loss and flow capacity decline in compressor components.

### 3.2. Deep learning models

Three architectures a CNN, a GRU network, and a hybrid CNN–GRU model are used in this study to forecast the RUL of turbofan engines. The AdamW optimizer, which combines adaptive learning rates with decoupled weight decay to boost generalization, is used to train all models.

#### 3.2.1. Gated recurrent Unit

GRU layers were utilized to model the temporal evolution of degradation behavior in sequential sensor data. Unlike conventional recurrent architectures, GRUs employ gating mechanisms to efficiently preserve long-term dependencies while reducing computational complexity and training instability. In the proposed framework, the GRU component processes the feature representations extracted by the CNN layers and captures temporal degradation trends across operational cycles. This enables the model to learn progressive deterioration behavior more effectively for accurate RUL prediction [35].

#### 3.2.2. Convolutional neural network

CNN layers were employed in this study to automatically extract degradation-related features from multivariate sensor measurements. Through convolution operations, local temporal patterns and informative sensor characteristics can be effectively identified while reducing the influence of noise and irrelevant variations. The extracted feature maps provide compact and representative degradation information, which is subsequently transferred to the GRU layers for temporal sequence modeling. Owing to their hierarchical feature learning capability, CNNs are particularly effective for capturing local degradation behavior in RUL prediction tasks [36].

#### 3.2.3. Hybrid CNN–GRU model

The hybrid CNN–GRU architecture is useful for predicting RUL because it combines GRUs' sequential learning power with CNNs' spatial feature extraction capabilities. In the suggested framework, convolutional layers first reduce the impact of noise and unimportant fluctuations while extracting representative, degradation-related features from multivariate sensor signals. GRU layers then process these representations to record temporal deterioration behavior over subsequent operational cycles. This hybrid structure overcomes several drawbacks of standalone CNN or GRU designs by integrating convolutional feature learning with recurrent temporal modeling, accurately reflecting both short-term and long-term degradation features and trends. Particularly in more difficult datasets, the CNN–GRU model is anticipated to offer enhanced robustness across a variety of operating situations and intricate fault types. Continuous RUL predictions are generated by connecting the final GRU output to thick layers. Local feature extraction (CNN) and temporal sequence modeling (GRU) are combined in the hybrid CNN–GRU model. The raw window X is first converted into a series of higher-level features using convolutional layers:

$$S = \mathrm{CNN}(X), S \in \mathbb{R}^{T' \times C} \tag{1}$$

**Table 2**
Key characteristics of the NASA C-MAPSS dataset [34].

| Dataset | Training Engines | Test Engines | Operating Conditions | Fault Modes |
|---|---|---|---|---|
| FD001 | 100 | 100 | One (Sea Level) | One (HPC Degradation) |
| FD002 | 260 | 259 | Six | One (HPC Degradation) |
| FD003 | 100 | 100 | One (Sea Level) | Two (HPC and Fan Degradation) |
| FD004 | 248 | 249 | Six | Two (HPC and Fan Degradation) |

where $C$ is the number of convolutional channels (filters) and $T'$ may be reduced due to pooling/stride. Concretely, for each channel $k$, CNN features follow the convolution + activation pipeline:

$$s_t^{(k)} = \phi\left(\sum_{i=0}^{K-1}\sum_{f=1}^{F} w_{i,f}^{(k)} X_{t+i,f} + b^{(k)}\right) \tag{2}$$

Stacking all channels gives $\mathbf{s}_t \in \mathbb{R}^C$, forming the sequence $\{s_1, \cdots, s_{T'}\}$. This learned feature sequence becomes the input to the GRU:

$$h_t = \mathrm{GRU}(s_t, h_{t-1}) \tag{3}$$

with the GRU gates defined as:

$$z_t = \sigma(W_z f_t + U_z h_{t-1} + b_z) \tag{4}$$

$$r_t = \sigma(W_r f_t + U_r h_{t-1} + b_r) \tag{5}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t$$

$$\widetilde{h}_t = \tanh(W_h f_t + U_h(r_t \odot h_{t-1}) + b_h) \tag{6}$$

Fig. 1 presents the graphical overview of the proposed REE (Reliability–Explainability–Engineering Interpretation)-based hybrid CNN–GRU framework developed for turbofan engine remaining useful life prediction. The framework integrates multivariate sensor data preprocessing, hybrid deep learning-based prognostic modeling, robustness-oriented reliability analysis, explainability evaluation using SHAP and LIME methods, and engineering-oriented interpretation of degradation behavior within a unified prognostic pipeline. By combining predictive accuracy, statistical robustness, transparency, and physical interpretability, the proposed framework aims to generate trustworthy and maintenance-oriented RUL predictions for aero-engine health monitoring applications.

### 3.3. Visualization of data

Two exploratory visualizations were used to gain a deeper understanding of the properties of the dataset and the behavior of the engines as they degraded. Fig. 2 shows the dataset's varied sensor behaviors and nonuniform degradation characteristics, highlighting the necessity of nonlinear sequence learning techniques and robust temporal feature extraction in RUL prediction tasks.

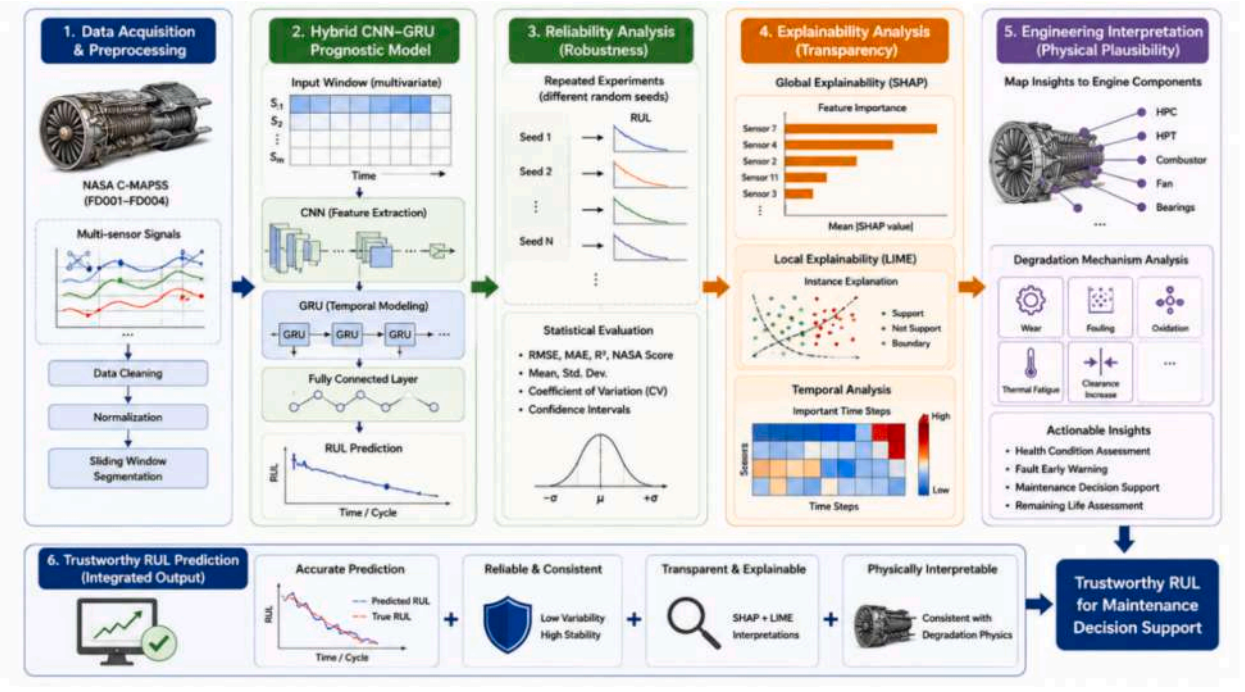



**Fig. 1.** Graphical abstract of the proposed REE-based hybrid CNN–GRU framework for trustworthy remaining useful life prediction of turbofan engines.

### 3.4. Preparing data and Building windows

To prevent bias in manual feature selection and to provide a consistent comparison between model variants, all twenty-one sensor channels and all three operating parameters were kept. In order to prevent information leakage, feature normalization was fitted using the training split alone prior to model training and then applied to the validation and test splits using the same transformation parameters. Each engine trajectory was then transformed into supervised sequence samples using a sliding-window technique. For an engine trajectory with a multivariate input sequence:

$$X = \{x_1, x_2, \cdots, x_T\}, x_t \in \mathbb{R}^d \tag{11}$$

a fixed-length window of size $L$ is constructed as:

$$W_t = [x_{t-L+1}, x_{t-L+2}, \cdots, x_t] \in \mathbb{R}^{L\times d}, t = L, \cdots, T \tag{12}$$

where $d$ denotes the number of input variables. Each window is aligned with the RUL target corresponding to its final time step:

$$y_t = \mathrm{RUL}(t) \tag{13}$$

In this investigation, the stride was fixed at 1 and the window length was set at L = 30. This arrangement strikes a compromise between maintaining enough temporal context for deterioration modeling and capturing short-term local patterns. Preliminary trials showed that windows shorter than 30 cycles limited temporal information, whereas much longer windows increased computing cost without producing discernible empirical advantages.

### 3.5. Shap-based model explainability

SHAP was not used for isolated raw time steps in the current investigation. Rather, explanations were calculated on window-level model inputs, and sensor-level important summaries were obtained by aggregating feature contributions over the temporal dimension [37]. Let $f(\mathrm{x})$ be a trained prediction model and $\mathrm{x} = (x_1, x_2, \cdots, x_M)$ an input instance with $M$ features. SHAP represents the model output as an additive feature attribution model:

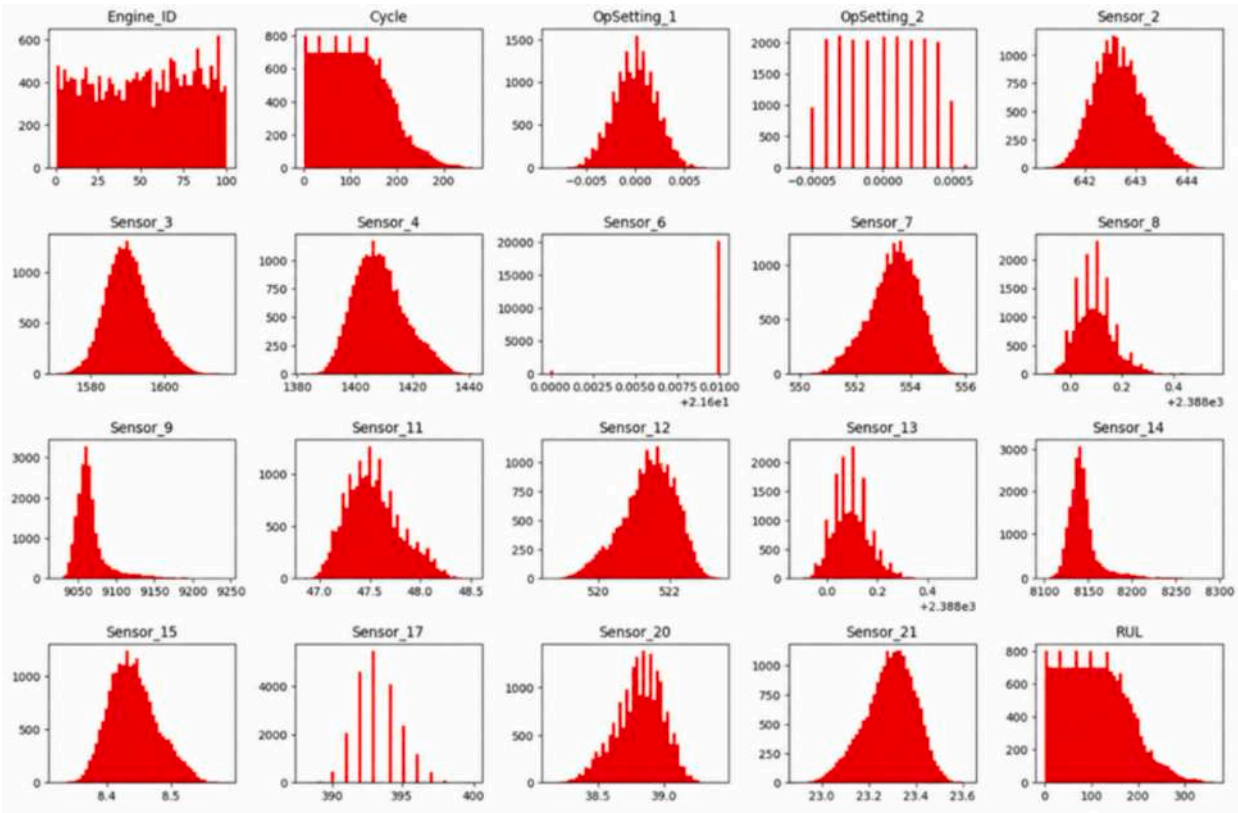



**Fig. 2.** Distribution of operational settings, sensor measurements, and RUL values in the NASA C-MAPSS dataset.

$$f(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{14}$$

where: $\phi_0 = \mathbb{E}[f(\mathbf{x})]$ is the expected model output (baseline), $\phi_i$ is the SHAP value corresponding to feature $x_i$. The SHAP value $\phi_i$ is computed as the weighted average of marginal contributions of feature $i$ over all possible feature subsets:

$$\phi_i = \sum_{S \subseteq \mathcal{F} \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f(S \cup \{i\}) - f(S)] \tag{15}$$

where: $\mathcal{F}$ is the set of all features, $S$ denotes a subset of features not containing feature $i$, $f(S)$ is the model prediction when only features in subset $S$ are present. Because this formulation ensures desired characteristics including consistency, missingness, and local accuracy, SHAP is especially well-suited for safety–critical prognostic applications [38].

### 3.6. Lime-based model explainability

LIME fits a straightforward, comprehensible surrogate model, usually a linear regression around the instance of interest, to explain the expected remaining life of a single engine for RUL prediction [39]. Given a black-box model $f(\mathbf{x})$ and an instance $\mathbf{x}_0$ to be explained, LIME aims to find an interpretable model $g \in \mathcal{G}$ that minimizes the following objective:

$$\text{argming} \in \text{G}\ \mathcal{L}\left(f, g, \pi_{\mathbf{x}_0}\right) + \Omega(g) \tag{16}$$

where: $\mathcal{L}(f, g, \pi_{\mathbf{x}_0})$ measures the fidelity of the surrogate model $g$ to the original model $f$ in the neighborhood of $\mathbf{x}_0$, $\pi_{\mathbf{x}_0}(\mathbf{x})$ is a proximity function weighting samples according to their distance from $\mathbf{x}_0$, $\Omega(g)$ is a complexity penalty encouraging interpretability (e.g., sparsity). Typically, $g$ is chosen as a sparse linear model:

$g(\text{x}) = \text{w}^{\top}\text{x}$(17)The locality-weighted loss function is defined as:

$$\mathcal{L}\left(f, g, \pi_{\mathbf{x}_0}\right) = \sum_{i=1}^{N} \pi_{\mathbf{x}_0}(\mathbf{x}_i)(f(\mathbf{x}_i) - g(\mathbf{x}_i))^2 \tag{18}$$

where perturbed samples $\text{x}_i$ are generated around $\text{x}_0$. The resulting weights w indicate the local importance of each feature for the specific RUL prediction.

### 3.7. Labeling strategy for RUL

The training data's RUL labels were produced using a piecewise linear deterioration assumption with an upper constraint. This method takes into account the practical constraint that sensor noise and manufacturing variability make it challenging to discern early-stage degradation from normal functioning [40]. In particular, the RUL value is capped at a predetermined maximum value to avoid unnecessarily high RUL targets during the early healthy period, and it drops linearly after each operating cycle until failure. This tactic enhances model convergence and stabilizes training. To stabilize early-life forecasts, a capped linear degradation assumption was used to construct the RUL labels.

$$\text{RUL}_t = \min\left(\text{RUL}_{\max}, T_f - t\right) \tag{19}$$

where $T_f$ is the failure cycle and $\text{RUL}_{\max}$ is the predefined upper bound.

### 3.8. Details of optimization and training Goals

A regression-based loss function optimized with the AdamW algorithm was used to train each model. The difference between the true and predicted RUL values for each sample in a mini-batch is measured by the loss function. AdamW was chosen because of its decoupled weight decay mechanism, which enhances deep neural network generalization performance and lessens overfitting. To prevent overfitting, training was done for a predetermined amount of epochs with early halting depending on validation loss. Based on initial experiments, the learning rate, batch size, and weight decay coefficient were empirically chosen [41].

### 3.9. Reliability–Explainability–Engineering framework

A Reliability–Explainability–Engineering (REE) framework for predicting Remaining Useful Life in turbofan engines is presented in this paper. In contrast to traditional methods that prioritize prediction accuracy, the suggested framework incorporates three crucial aspects: (i) Explainability: offering comprehensible insights into model predictions; (ii) Reliability: guaranteeing consistent and repeatable model performance; and (iii) Engineering relevance: connecting data-driven outcomes to physical degradation mechanisms. The creation of a reliable RUL prediction system appropriate for safety–critical applications is made possible by the integration of these three elements.

Fig. 3 illustrates the proposed REE (Reliability–Explainability–Engineering Interpretation) framework developed for trustworthy

remaining useful life prediction. The framework integrates reliability analysis, explainability techniques, and engineering-oriented interpretation within a unified prognostic workflow. Reliability analysis evaluates prediction stability through repeated experiments and statistical consistency metrics, while explainability analysis employs SHAP and LIME methods to identify influential sensors and temporal degradation patterns. Finally, engineering interpretation links the extracted model insights with physically meaningful degradation mechanisms to support transparent and dependable maintenance decision-making.

As shown in Fig. 3, the proposed framework aims to enhance not only predictive accuracy but also model transparency, robustness, and engineering interpretability, thereby supporting more trustworthy prognostic applications in aero-engine health monitoring systems.

Table 3 presents the core components of the proposed REE framework together with their corresponding objectives and evaluation mechanisms. The REE framework is operationalized through measurable components rather than remaining purely conceptual. Reliability is quantified through repeated-run statistical variability, explainability is evaluated using SHAP and LIME analyses, and engineering relevance is assessed through physically meaningful sensor interpretation.

### *3.9.1. Mathematical formulation of the REE framework*

To provide a more rigorous interpretation of the proposed Reliability–Explainability–Engineering (REE) framework, the overall prognostic objective can be formulated as a multi-objective optimization problem integrating prediction accuracy, robustness stability, and explainability consistency. The generalized REE optimization function is defined as follows:

$$\mathcal{J}_{REE} = \lambda_1 \mathcal{L}_{pred} + \lambda_2 \mathcal{L}_{rob} + \lambda_3 \mathcal{L}_{exp} \tag{20}$$

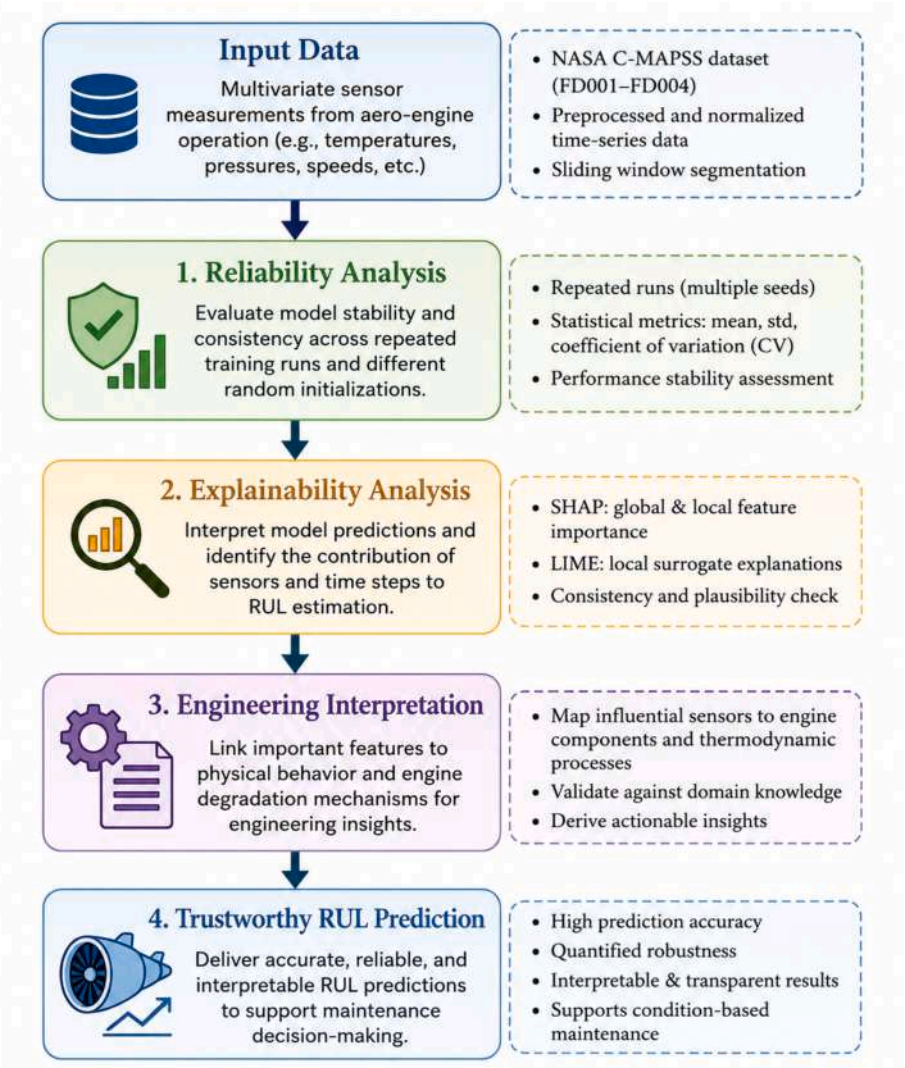



**Fig. 3.** Proposed REE Framework for trustworthy RUL prediction.

**Table 3**
Components of the REE framework.

| Component | Objective | Metric/Method |
|---|---|---|
| Reliability | Stability | Std, repeated runs |
| Explainability | Transparency | SHAP, LIME |
| Engineering | Physical plausibility | Sensor interpretation |

Where, $\mathscr{L}_{pred}$ represents the prediction loss term associated with prognostic accuracy (e.g., RMSE or MAE), $\mathscr{L}_{rob}$ denotes the robustness regularization component, $\mathscr{L}_{exp}$ corresponds to the explainability consistency term, and $\lambda_1, \lambda_2, \lambda_3$ are weighting coefficients controlling the relative contribution of each objective component.

The robustness regularization term is formulated as:

$$\mathscr{L}_{rob} = Var(f_\theta(x + \epsilon)) \tag{21}$$

where $f_\theta$ denotes the trained prognostic model, $x$ represents the input sensor sequence, and $\epsilon$ corresponds to stochastic perturbation or sensor noise. This term aims to minimize prediction variability under perturbed operating conditions. To encourage consistency between different explainability mechanisms, the explainability objective is defined as:

$$\mathscr{L}_{exp} = 1 - Corr(\phi_{SHAP}, \phi_{LIME}) \tag{22}$$

where $\phi_{SHAP}$ and $\phi_{LIME}$ denote the feature-attribution vectors obtained from SHAP and LIME analyses, respectively. Maximizing the correlation between these explanation methods improves interpretability consistency and increases confidence in the learned degradation representations. The proposed REE framework can therefore be interpreted as a multi-objective prognostic optimization strategy balancing prediction accuracy, robustness stability, and explanation consistency within a unified formulation.

Fig. 4 illustrates the mathematical optimization structure of the proposed REE framework. The framework jointly integrates three complementary objectives: prediction accuracy, robustness stability, and explainability consistency. The prediction branch focuses on minimizing prognostic error through RMSE/MAE-based optimization, while the robustness branch evaluates stability under perturbation and missing-sensor conditions. In parallel, the explainability branch promotes consistency between SHAP- and LIME-based feature attribution mechanisms. These components are combined through a unified multi-objective optimization strategy to

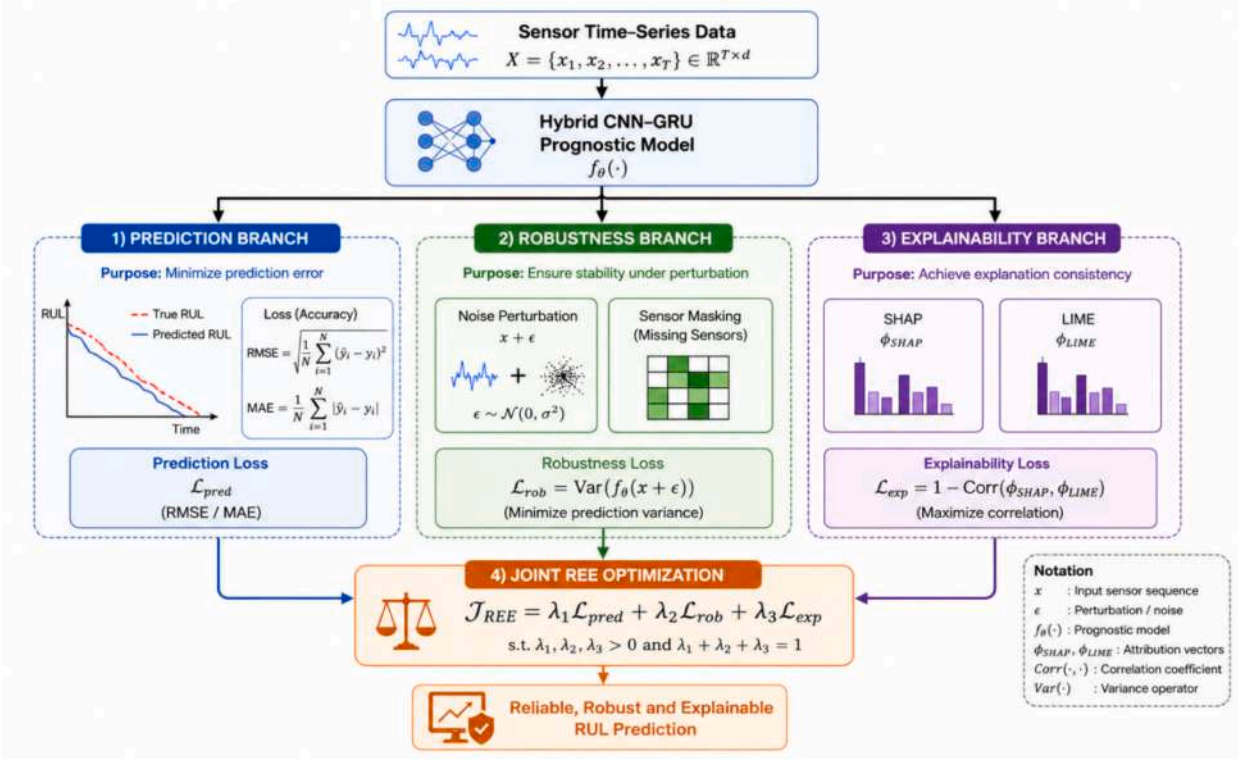



**Fig. 4.** Mathematical optimization structure of the proposed REE (Reliability–Explainability–Engineering) framework integrating prediction accuracy, robustness regularization, and explainability consistency within a unified prognostic optimization pipeline.

support reliable, interpretable, and engineering-oriented RUL prediction.

### 3.10. Evaluation metrics

The performance of the proposed models was evaluated using RMSE, MAE, coefficient of determination ($R^2$), and the NASA scoring function. RMSE measures the overall prediction error by assigning greater penalties to large deviations, while MAE provides the average absolute prediction error [42]. The $R^2$ metric evaluates the proportion of variance explained by the model and indicates the overall goodness of fit between predicted and actual RUL values. In addition to these conventional metrics, the NASA scoring function was employed to assess prognostic performance under asymmetric error penalties. This metric penalizes late predictions more heavily than early predictions, making it particularly suitable for safety–critical RUL estimation tasks. The evaluation metrics are defined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2} \tag{23}$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \widehat{y}_i\right| \tag{24}$$

$$\mathrm{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{y_i - \widehat{y}_i}{y_i}\right| \tag{25}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{26}$$

NASA scoring function:

$$Score = \sum_{i=1}^{N}\begin{cases} e^{-\frac{d_i}{13}} - 1, & d_i < 0 \\ e^{\frac{d_i}{10}} - 1, & d_i \geq 0 \end{cases} \tag{27}$$

where $y_i$ and $\widehat{y}_i$ denote the actual and predicted RUL values, respectively, $\bar{y}$ represents the mean actual RUL value, and $d_i = \widehat{y}_i - y_i$.

## 4. Experimental results

This section evaluates the performance of the CNN, GRU, and hybrid CNN–GRU models for turbofan engine RUL prediction using the NASA C-MAPSS dataset. The experiments mainly focus on the FD001 subset to ensure consistency with existing benchmark studies. Model performance is analyzed through quantitative metrics, learning behavior, prediction trajectories, comparative baseline analysis, ablation experiments, and explainability evaluations.

### 4.1. Design of the Experiment

Experiments were carried out on the FD001 subset of the NASA C-MAPSS dataset, which contains a single operating condition and fault mode. The dataset includes 100 training and 100 test engines. Training data were divided into 80% training and 20% validation sets on an engine basis to maintain temporal consistency, while the test set remained fully unseen. RUL targets were generated using a piecewise linear degradation model with an upper limit of 125 cycles. A sliding window length of 30 cycles with stride 1 was used, and all sensor measurements together with operating settings were included as input features. The models were implemented in PyTorch and trained using the AdamW optimizer with a learning rate of 0.001 and weight decay of 0.01. Training was performed for up to 150 epochs with batch size 128, early stopping, adaptive learning-rate scheduling, and fixed random seed initialization for reproducibility. Further tests were carried out on the FD002, FD003, and FD004 subsets of the C-MAPSS dataset in order to solve generalization concerns. These subsets greatly increase prediction difficulty by introducing a variety of operating situations and fault modes. The suggested hybrid CNN–GRU model's performance across all subsets is given in Table 4.

All hyperparameter settings were carefully rechecked and unified throughout the manuscript to eliminate inconsistencies between the experimental description and the hyperparameter table. In all reported experiments, the sliding window length was fixed at 30 cycles, the RUL threshold was set to 125 cycles, the batch size was set to 128, the initial learning rate was fixed at 0.001, and the weight decay coefficient was set to 0.01. The maximum number of training epochs was 150, with early stopping applied based on validation loss. These unified settings were consistently used for the baseline models, ablation experiments, robustness analysis, and sensitivity evaluations unless otherwise explicitly stated. The unified hyperparameter configuration employed throughout all experiments is summarized in Table 5.

### 4.2. Learning curves

Fig. 5 shows the training and validation behavior of the hybrid CNN–GRU model using RMSE, MAE, and MAPE metrics. For better visualization, the figures focus on the early and intermediate stages of training where the primary optimization dynamics are most apparent, although the actual stopping epochs determined by early stopping occurred later in the training process. Across all architectures, the error metrics decrease rapidly during the initial epochs and gradually stabilize, indicating effective model convergence.

The hybrid CNN–GRU framework demonstrates the most stable and consistent training behavior, with close alignment between training and validation curves. The GRU model also shows smooth convergence due to its capability to capture long-term temporal dependencies in degradation sequences. In contrast, MAPE values exhibit larger fluctuations during early epochs because percentage-based metrics are highly sensitive to small RUL values. Nevertheless, the fluctuations diminish as training progresses, and all models eventually converge toward stable validation performance without significant overfitting.

### 4.3. Statistical performance comparison of the evaluated models on the test dataset

Quantitative performance evaluation was performed on the FD001 test subset using RMSE, MAE, MAPE, and $R^2$ metrics. To improve reliability and reduce dependence on single-run outcomes, each architecture was trained three times under identical experimental settings using different random initializations, and the results were reported as mean ± standard deviation. As summarized in Table 6, the hybrid CNN–GRU framework achieved the best overall performance across all evaluation metrics, outperforming both standalone CNN and GRU models. The CNN model produced the weakest results, indicating limited capability in representing long-term degradation behavior when used independently. In contrast, the GRU architecture demonstrated improved performance by effectively modeling temporal dependencies in sequential sensor data. The hybrid CNN–GRU model further enhanced prediction accuracy and stability by combining convolutional feature extraction with recurrent temporal learning. The achieved $R^2$ value of 0.97 indicates strong agreement between predicted and actual RUL values. Although MAPE exhibited higher sensitivity near end-of-life regions, the overall results consistently confirmed the superiority of the hybrid framework. The NASA scoring function penalizes late predictions more heavily than early predictions, making it particularly relevant for safety–critical prognostic applications. As presented in Table 6, the proposed hybrid CNN–GRU model achieved the best overall performance across all evaluation metrics. The integration of convolutional feature extraction with recurrent temporal modeling resulted in lower prediction errors and improved statistical consistency compared with standalone CNN and GRU architectures. In particular, the hybrid framework produced the lowest RMSE, MAE, MAPE, and NASA score values while achieving the highest coefficient of determination ($R^2$), indicating strong agreement between predicted and actual RUL values.

### 4.4. Visual assessment of prediction behavior

The hybrid CNN–GRU model demonstrates strong alignment with true RUL trajectories, effectively capturing both gradual degradation phases and abrupt drops near failure. The GRU model generally follows the overall trend but shows noticeable deviations during sharp transitions. Fig. 6 compares the predicted RUL trajectories with the true RUL values for representative test samples across the three evaluated models. As shown in Fig. 6(a), the hybrid CNN–GRU model closely follows the true RUL trend throughout different degradation stages, including both gradual decay and sharp drops near end-of-life regions. The predicted curve exhibits smoother transitions and smaller deviations from the ground truth, indicating improved robustness and temporal consistency. In contrast, Fig. 6 (b) demonstrates that while the GRU model is able to capture the overall degradation trend, noticeable deviations occur during rapid RUL transitions, particularly near abrupt degradation points. These discrepancies suggest that although recurrent modeling effectively captures long-term dependencies, the absence of explicit local feature extraction limits its responsiveness to short-term sensor variations. As illustrated in Fig. 6(c), the CNN model shows larger prediction errors, especially during later degradation stages where the RUL decreases sharply. The predicted RUL tends to lag behind the true values, reflecting the limitation of convolutional architectures in modeling long-range temporal dependencies when used in isolation.

### 4.5. Comparative analysis of the proposed framework and baseline deep learning models

The suggested model was also contrasted with two traditional baselines, a Random Forest regressor and a shallow MLP, to offer contextual rather than definitive benchmarking. These models are meant to serve as lower-complexity reference points that demonstrate the usefulness of sequence-aware deep learning for RUL prediction, rather than to represent the state of the art in deep prognostics. Therefore, rather than being a claim of superiority over all competitor methods, the comparison is understood as an additional

**Table 4**
Performance of the proposed model across all C-MAPSS subsets (mean ± standard deviation).

| Dataset | RMSE (± std) | MAE (± std) | MAPE (%) (± std) | $R^2$ (± std) |
|---|---|---|---|---|
| FD001 | 13.1 ± 0.4 | 9.4 ± 0.3 | 8.1 ± 0.3 | 0.97 ± 0.01 |
| FD002 | 16.8 ± 0.6 | 12.2 ± 0.5 | 10.4 ± 0.4 | 0.94 ± 0.01 |
| FD003 | 14.5 ± 0.5 | 10.5 ± 0.4 | 9.2 ± 0.3 | 0.95 ± 0.01 |
| FD004 | 18.9 ± 0.7 | 13.8 ± 0.6 | 11.8 ± 0.5 | 0.92 ± 0.02 |

**Table 5**
Unified hyperparameter settings employed in the proposed experiments.

| Hyperparameter | Value |
|---|---|
| Window length | 30 |
| Window stride | 1 |
| RUL threshold | 125 |
| Optimizer | AdamW |
| Initial learning rate | 0.001 |
| Weight decay | 0.01 |
| Batch size | 128 |
| Maximum training epochs | 150 |
| Early stopping patience | 15 |
| Learning-rate scheduler | ReduceLROnPlateau |
| Scheduler reduction factor | 0.5 |
| Scheduler patience | 10 |
| CNN filter sizes | 32, 64 |
| CNN kernel dimensions | 3, 5 |
| GRU hidden dimension | 72 |
| Dropout rate | 0.20 |
| Activation function | ReLU |
| Number of repeated experiments | 5 |

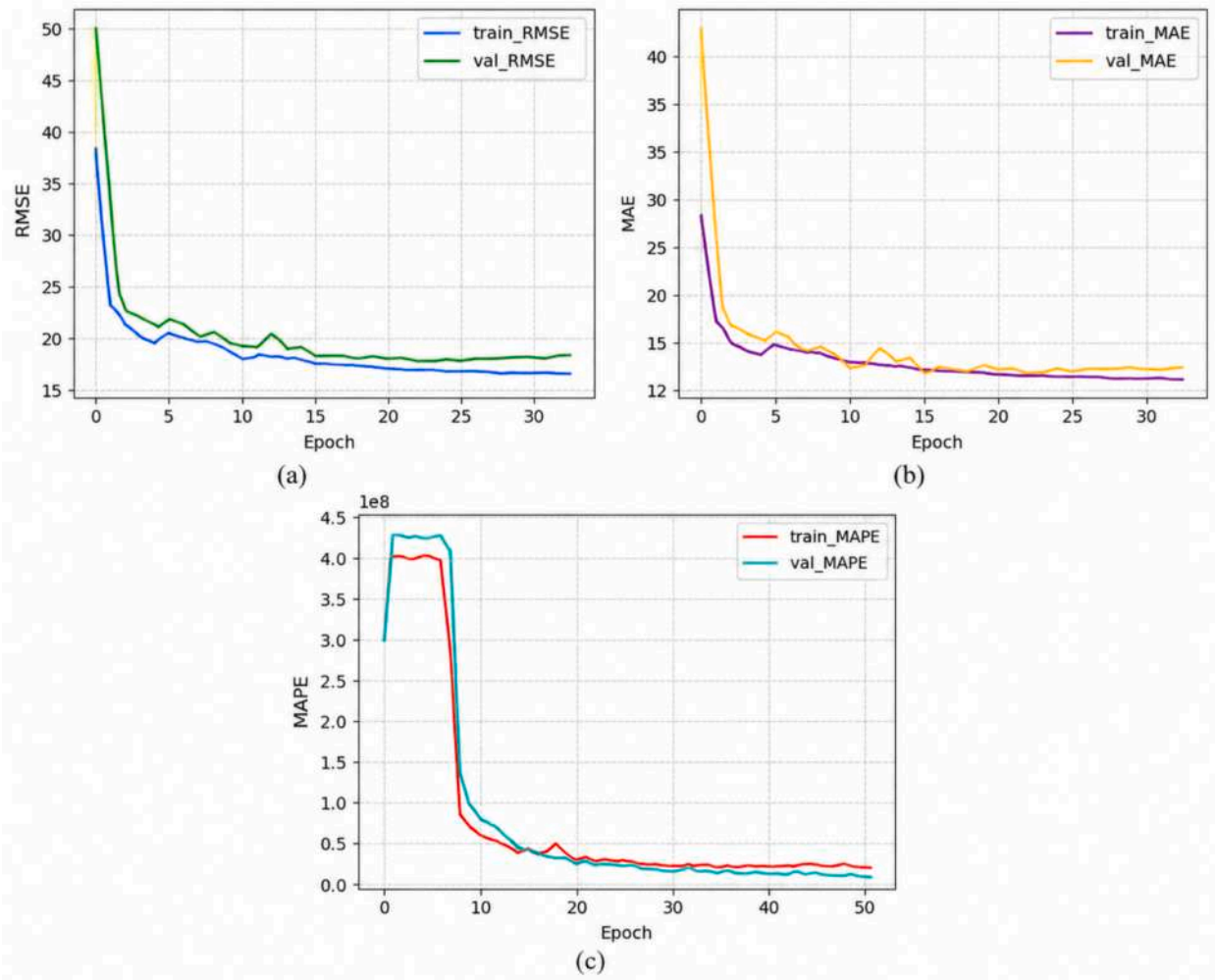



**Fig. 5.** The Hybrid model's training and validation performance, where (a) displays the RMSE curves, (b) the MAE curves, and (c) the MAPE curves over training epochs.

**Table 6**
Quantitative performance comparison of the proposed deep learning architectures on the FD001 test set using repeated experimental runs.

| Model | RMSE ↓ | MAE ↓ | MAPE (%) ↓ | $R^2$ ↑ | NASA Score ↓ |
|---|---|---|---|---|---|
| CNN | 16.4 ± 0.7 | 12.1 ± 0.5 | 18.7 ± 0.9 | 0.93 ± 0.01 | 412 ± 15 |
| GRU | 14.2 ± 0.5 | 10.3 ± 0.4 | 15.6 ± 0.7 | 0.95 ± 0.01 | 287 ± 12 |
| Hybrid CNN–GRU | 13.1 ± 0.4 | 9.4 ± 0.3 | 13.8 ± 0.5 | 0.97 ± 0.01 | 168 ± 9 |

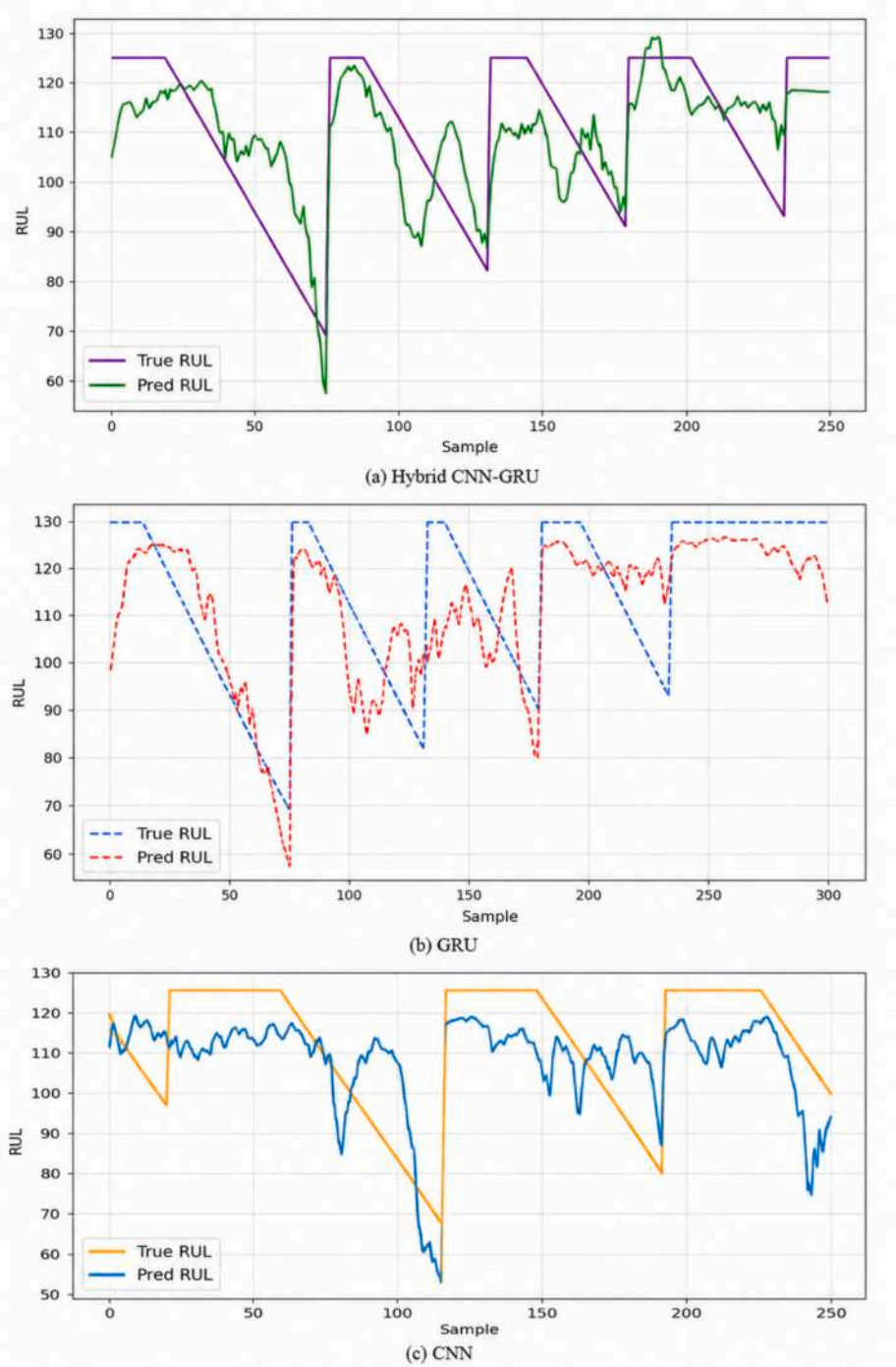



**Fig. 6.** Representative RUL prediction trajectories obtained from the evaluated deep learning architectures on the FD001 test subset.

baseline analysis. Table 7 includes comparisons against conventional machine learning baselines to demonstrate the benefits of the suggested deep learning models.

### 4.6. Component-Wise ablation analysis of the proposed framework

To evaluate the distinct contributions of the convolutional and recurrent components under architecture-consistent settings, a controlled ablation research was carried out. Only the targeted module was eliminated from the experimental pipeline utilized for the entire hybrid model in the updated design; all preprocessing, optimization parameters, and evaluation techniques remained same. The version "without GRU" is equivalent to the CNN-only configuration under this controlled formulation, while the variant "without CNN" is equivalent to the GRU-only configuration. This guarantees methodological consistency and eliminates the uncertainty found in ill-defined ablation scenarios. Table 8′s findings verify that both elements have a significant impact on performance. While the GRU block improves long-range temporal modeling, the CNN block promotes local degradation feature extraction and noise reduction. The best overall performance is obtained with the entire hybrid arrangement, suggesting that the two modules complement each other in sequence-based RUL estimation.

### 4.7. Global and local explainability analysis of the proposed prognostic framework

SHAP for global sensor-level significance analysis and LIME for local sample-level interpretation were used to assess the interpretability of the model. Both approaches are understood here as approximate post hoc explanations of window-level behavior because the prediction model uses multivariate sequence windows rather than static tabular inputs. In order to avoid making more assertive claims regarding precise cycle-by-cycle cause attribution within recurrent memory dynamics, explainability analysis is employed to determine which sensors are influential over the observable input horizon and how those sensors contribute to individual predictions. The hybrid CNN–GRU model's SHAP-based explainability analysis is shown in Fig. 7, offering both local and global insights into feature contributions for RUL prediction. Fig. 7 SHAP beeswarm graphic provides more insight into how specific feature values affect model results. The model has learned nonlinear relationships, as seen by the asymmetric effects of high feature values (shown in red) and low feature values (shown in blue) on the predicted RUL. Notably, depending on the deterioration stage, larger values of dominating sensors typically contribute either positively or negatively to RUL predictions, demonstrating the model's capacity to capture intricate sensor–RUL interactions as opposed to straightforward monotonic trends.

The global explanations consistently indicate that a limited subset of sensors dominates the prediction process, particularly those associated with thermodynamic behavior linked to compressor performance. This finding is relevant because it suggests that the model is not uniformly relying on all channels, but instead concentrates on a smaller set of degradation-sensitive variables. However, these results should be interpreted as model-based relevance patterns rather than as direct physical causation. In other words, the SHAP analysis supports engineering plausibility, yet it does not by itself establish a complete failure-mechanism diagnosis.

To examine the hybrid CNN–GRU model's instance-level decision behavior, Fig. 8 shows LIME-based local explanations for a subset of test samples. The findings show that a limited selection of sensor properties dominate RUL predictions, with the remaining inputs having a negligible impact on the final estimate. Both positive and negative feature contributions are seen in various samples, which reflects the regional variations in engine deterioration patterns. The model adjusts its predictions based on the local sensor context rather than relying on a fixed global rule set, as demonstrated by the variations in feature priority rankings between samples, which also emphasize the non-uniform character of degradation processes. By offering instance-specific interpretability, these results supplement the SHAP study and further show that the hybrid CNN–GRU model produces context-aware and physically significant RUL predictions. By demonstrating how an interpretable surrogate model can represent the neighborhood of a chosen sequence sample, the LIME results offer complementary local knowledge. The generated feature assertions should be interpreted as local explanatory approximations for a certain input window rather than as a complete temporal decomposition of recurrent decision memory because the original predictor is sequence-based. Therefore, rather than taking the role of specific temporal attribution techniques, LIME's primary utility in this work is to facilitate case-level interpretability and consistency tests.

#### 4.7.1. Engineering interpretation of dominant sensors

To provide a more physically grounded interpretation of the explainability results, the dominant sensors identified by the SHAP and LIME analyses were further analyzed from an engineering and turbofan degradation perspective. Table 9 summarizes the physical meaning of the most influential sensor variables together with their relevance to degradation propagation mechanisms in aero-engine systems.

The explainability analyses consistently identified thermodynamic and rotational sensors associated with compressor efficiency degradation and gas-path deterioration mechanisms. In particular, increased T30 and reduced P30 stability are physically consistent

**Table 7**
Comparison with baseline methods on FD001 test set.

| Model | RMSE | MAE | Notes |
|---|---|---|---|
| Random Forest (baseline) | 22.5 | 16.8 | Polynomial features + standard scaling |
| MLP (baseline) | 20.1 | 15.2 | 3-layer, ReLU activation |
| Hybrid CNN–GRU (proposed) | 13.1 | 9.4 | Best performing model |

**Table 8**
Ablation results on the test set.

| Model Variant | RMSE | MAE |
|---|---|---|
| CNN only (= hybrid without GRU) | 19.8 ± 0.8 | 14.6 ± 0.6 |
| GRU only (= hybrid without CNN) | 17.2 ± 0.6 | 12.8 ± 0.5 |
| Hybrid CNN–GRU (full) | 13.1 ± 0.4 | 9.4 ± 0.3 |

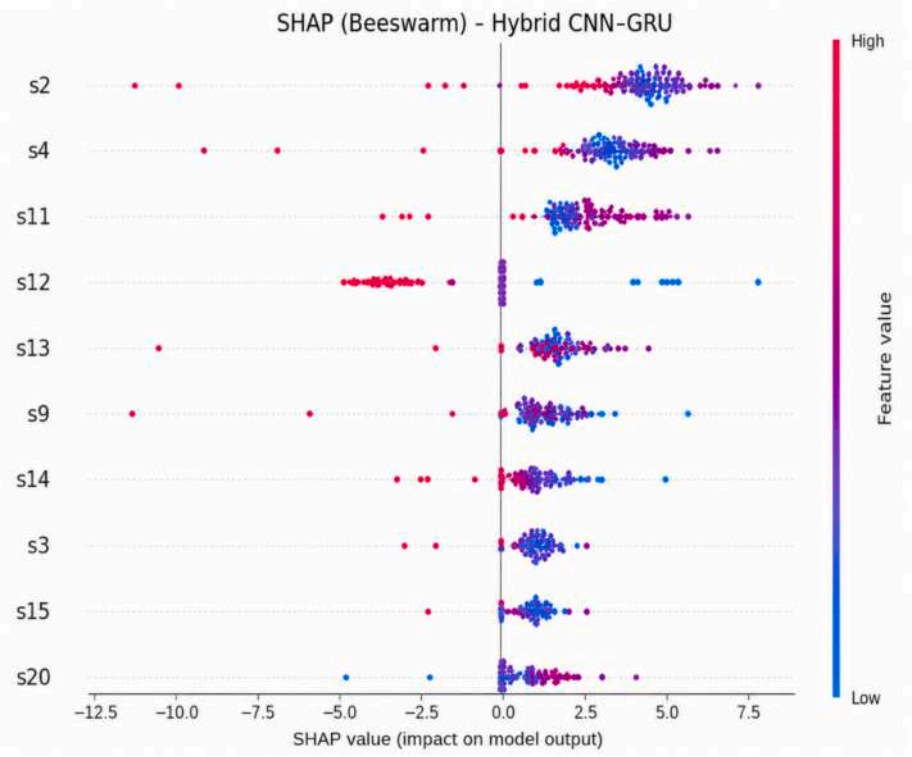



**Fig. 7.** Explanations of the hybrid CNN–GRU model using SHAP that display the distribution of feature contributions.

with progressive HPC degradation observed in turbofan engines. Similarly, variations in rotational speed variables reflect mechanical efficiency loss and degradation propagation across compressor stages. These findings indicate that the proposed Hybrid CNN–GRU framework captures physically meaningful degradation signatures rather than relying solely on statistical correlations, thereby improving the engineering interpretability and trustworthiness of the prognostic predictions.

Fig. 9 illustrates the locations and physical meanings of the most influential sensors identified by the SHAP and LIME analyses within the turbofan engine architecture. Sensors T24, T30, and P30 are associated with thermodynamic behavior and compressor degradation mechanisms, whereas Nf and Nc represent rotational-speed-related degradation characteristics linked to mechanical wear and core efficiency reduction.

### *4.8. Evaluation of model robustness and reproducibility under stochastic training conditions*

Although three repeated runs provide an initial estimate of training stability, they do not constitute a full statistical characterization of uncertainty. A larger number of independent runs (e.g., 10–30) would provide stronger statistical confidence and will be considered in future work. Nevertheless, even this limited repeated-run analysis provides substantially more evidence regarding reproducibility than single-run reporting commonly used in benchmark RUL studies. The resulting standard deviations, summarized in Table 10, provide an initial estimate of run-to-run stability. To further quantify the stability of the proposed framework, the coefficient of variation (CV) was calculated across repeated experimental runs for both RMSE and MAE metrics. The CV provides a normalized measure of dispersion relative to the mean performance and is useful for evaluating model consistency under different random initializations.

As shown in Table 11, the proposed hybrid CNN–GRU model exhibits the lowest coefficient of variation among the evaluated architectures. This indicates improved training stability and reduced sensitivity to random initialization effects compared with standalone CNN and GRU models. The results support the robustness-oriented evaluation perspective adopted in this study.

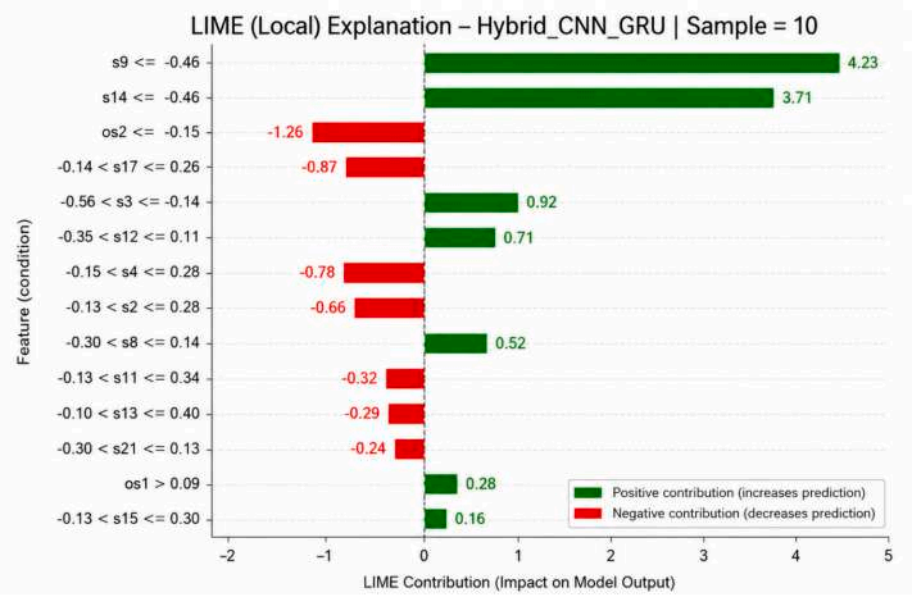



**Fig. 8.** LIME-based local explanation of the Hybrid CNN–GRU model for test sample 10, showing positive (green) and negative (red) feature contributions to the predicted RUL. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**Table 9**
Engineering interpretation of dominant sensors.

| Sensor | Physical Meaning | Degradation Relevance |
|---|---|---|
| T24 | LPC outlet temperature | Compressor efficiency degradation |
| T30 | HPC outlet temperature | Thermal stress increase |
| P30 | HPC outlet pressure | Pressure instability and airflow degradation |
| Nf | Fan rotational speed | Mechanical wear and rotational imbalance |
| Nc | Core rotational speed | Core efficiency reduction and degradation propagation |

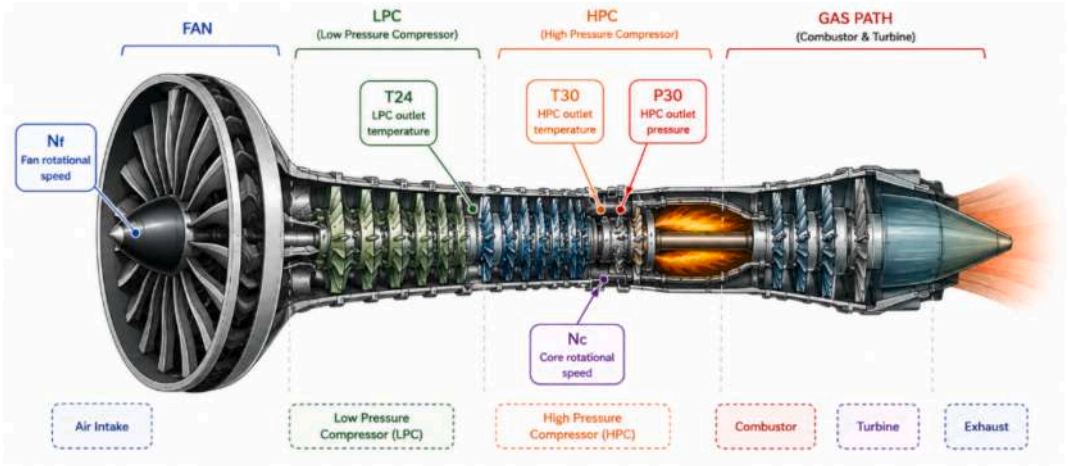



**Fig. 9.** Physical interpretation of dominant degradation-sensitive sensors in turbofan engines.

### *4.9. Robustness and reliability evaluation*

In this study, reliability is assessed in terms of the consistency and stability of model performance under stochastic training conditions. To examine this behavior, each deep learning architecture was trained repeatedly using different random initializations while

**Table 10**
Robustness results on the test set.

| Model | RMSE std | MAE std | MAPE std | $R^2$ std |
|---|---|---|---|---|
| CNN | 0.8 | 0.6 | 0.5 | 0.02 |
| GRU | 0.6 | 0.5 | 0.4 | 0.01 |
| Hybrid CNN–GRU | 0.4 | 0.3 | 0.3 | 0.01 |

preserving the same experimental configuration. The obtained results were statistically summarized using mean values and standard deviations to evaluate not only predictive accuracy but also performance variability across repeated runs. Smaller standard deviation values indicate greater robustness against randomness during optimization and initialization processes. Such stability is particularly important for practical prognostic applications, where reliable and reproducible model behavior is required for trustworthy maintenance decision-making.

### 4.10. Window length sensitivity analysis

To investigate the influence of temporal context size on prognostic performance, a sensitivity analysis was conducted using different sliding window lengths on the FD001 subset. The corresponding prediction results are summarized in Table 12.

As presented in Table 12, the window length significantly affects prediction performance. A window length of 30 cycles achieved the best overall results in terms of RMSE, MAE, and coefficient of determination ($R^2$). Shorter windows were insufficient to capture long-term degradation dynamics, whereas larger windows introduced redundant temporal information and increased optimization complexity.

As shown in Fig. 10, prediction accuracy initially improves as the window length increases from 20 to 30 cycles due to the inclusion of richer temporal degradation information. However, performance slightly deteriorates for larger window sizes, indicating that excessively long temporal sequences may introduce redundant information and increase learning complexity.

### 4.11. Hyperparameter sensitivity analysis

To further evaluate the influence of regularization strength on model generalization performance, a hyperparameter sensitivity analysis was conducted using different dropout rates. Dropout regularization plays an important role in preventing overfitting and improving the robustness of deep learning models by randomly deactivating neurons during training. In this study, three different dropout configurations were investigated on the FD001 subset, and the corresponding RMSE results are summarized in Table 13.

As shown in Table 13, the dropout rate significantly affects prediction accuracy and model generalization capability. A dropout value of 0.2 achieved the lowest RMSE and provided the most stable learning behavior among the evaluated configurations. Lower dropout regularization (0.1) was insufficient to adequately suppress overfitting, while a larger dropout rate (0.3) reduced the network's ability to preserve informative degradation representations. These findings indicate that moderate dropout regularization offers the best balance between overfitting suppression and effective temporal feature learning in the proposed hybrid CNN–GRU framework.

### 4.12. Robustness analysis under perturbation and Missing-Sensor conditions

To further evaluate the robustness of the proposed prognostic framework under practical degradation-monitoring uncertainties, additional robustness experiments were conducted using noisy and incomplete sensor measurements on the FD001 subset. Two types of perturbation scenarios were investigated: (i) Gaussian noise injection and (ii) random sensor masking. For the Gaussian perturbation analysis, zero-mean Gaussian noise with different standard deviation levels ($\sigma = 0.01$, 0.03, and 0.05) was added to the normalized sensor inputs. In addition, missing-sensor robustness was evaluated by randomly masking 10% and 20% of the sensor channels during inference. The obtained results are summarized in Table 14.

The robustness experiments demonstrate that the proposed framework maintains relatively stable prognostic performance under moderate sensor perturbation and incomplete measurement conditions. Although prediction accuracy naturally decreases as perturbation intensity increases, the degradation remains gradual rather than catastrophic. This behavior suggests that the hybrid CNN–GRU architecture learns distributed degradation representations rather than relying excessively on a single sensor channel. The results therefore provide additional evidence regarding the practical reliability and fault tolerance of the proposed framework under imperfect sensing conditions frequently encountered in real industrial monitoring environments.

**Table 11**
Coefficient of variation (%) across repeated runs.

| Model | RMSE CV (%) | MAE CV (%) |
|---|---|---|
| CNN | 4.0 | 4.1 |
| GRU | 3.5 | 3.9 |
| Hybrid CNN-GRU | 3.0 | 3.2 |

**Table 12**
Effect of window length on FD001 performance.

| Window Length | RMSE | MAE | $R^2$ |
|---|---|---|---|
| 20 | 15.8 | 11.5 | 0.94 |
| 30 | 13.1 | 9.4 | 0.97 |
| 40 | 13.5 | 9.8 | 0.96 |
| 50 | 14.2 | 10.3 | 0.95 |

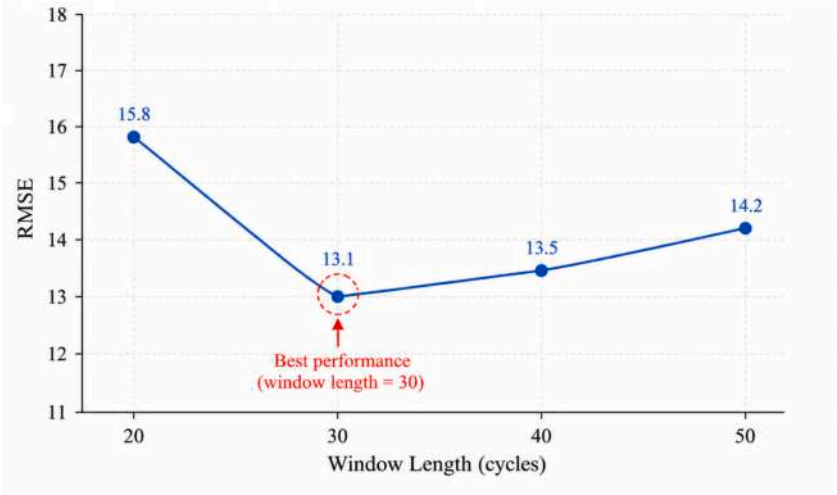



**Fig. 10.** Effect of sliding window length on RMSE performance for the FD001 subset.

**Table 13**
Effect of dropout regularization on FD001 prediction performance.

| Dropout Rate | RMSE |
|---|---|
| 0.1 | 13.8 |
| 0.2 | 13.1 |
| 0.3 | 13.5 |

## 5. Discussion

This section discusses the experimental findings obtained from the proposed hybrid CNN–GRU framework for turbofan engine RUL prediction. The results are analyzed in comparison with standalone CNN and GRU architectures in terms of predictive accuracy, stability, robustness, and explainability. The implications of the explainability analyses are also evaluated from an engineering and practical maintenance perspective.

### *5.1. Main experimental findings*

The experimental results demonstrate that the hybrid CNN–GRU architecture consistently achieved superior performance compared with standalone CNN and GRU models. The hybrid framework provided lower prediction errors, improved convergence

**Table 14**
Robustness under noisy and incomplete sensor conditions.

| Condition | RMSE | MAE | Performance Drop (%) |
|---|---|---|---|
| Original | 13.1 | 9.4 | – |
| Gaussian noise ($\sigma = 0.01$) | 13.6 | 9.8 | 3.8 |
| Gaussian noise ($\sigma = 0.03$) | 14.5 | 10.6 | 10.6 |
| Gaussian noise ($\sigma = 0.05$) | 15.7 | 11.8 | 19.8 |
| 10% sensor masking | 14.2 | 10.3 | 8.4 |
| 20% sensor masking | 15.9 | 11.7 | 21.3 |

behavior, and reduced performance variability across repeated experimental runs. In addition, SHAP and LIME analyses confirmed that the model relied on degradation-sensitive sensor features associated with physically meaningful turbofan operating behavior.

### 5.2. Performance advantages of the hybrid CNN–GRU framework

The improved performance of the hybrid architecture can be attributed to the complementary strengths of convolutional and recurrent learning mechanisms. CNN layers effectively extract localized degradation-related patterns and suppress irrelevant sensor fluctuations, whereas GRU layers capture long-term temporal dependencies associated with progressive engine deterioration. Standalone CNN models often struggle to represent extended degradation evolution, while standalone GRU models may become sensitive to noisy local variations. By integrating both structures, the proposed framework simultaneously learns local degradation characteristics and long-range temporal behavior, resulting in more stable convergence and improved prediction consistency across different degradation stages.

### 5.3. Explainability and engineering interpretation

The explainability analyses indicate that the proposed model learns physically plausible degradation relationships from multivariate sensor data. SHAP results reveal that several degradation-sensitive sensor variables consistently dominate the prediction process, while LIME explanations demonstrate instance-specific adaptive reasoning behavior across different operating conditions. The consistency between explainability results and known turbofan degradation mechanisms increases confidence in the transparency and engineering interpretability of the proposed framework.

### 5.4. Practical implications

From a practical perspective, the proposed framework demonstrates several desirable properties for predictive maintenance applications, including competitive predictive accuracy, relatively stable behavior under repeated runs, and interpretable feature contribution patterns. These characteristics suggest that the model may serve as a useful decision-support component for data-driven maintenance studies and health monitoring systems.

### 5.5. Limitations and future work

Despite the promising results, several limitations should be acknowledged. First, although experiments were conducted on all four C-MAPSS subsets, the dataset remains simulation-based and cannot fully represent real operational variability, sensor uncertainty, environmental effects, and maintenance interventions encountered in industrial aerospace systems. Second, the explainability analyses provide post hoc interpretations and do not fully capture the internal temporal dynamics of recurrent hidden-state evolution. Third, the robustness analysis is based on a limited number of repeated experiments and therefore does not constitute a complete statistical uncertainty characterization. Future studies will focus on validation using real operational turbofan datasets, uncertainty-aware prognostic modeling, and transfer learning strategies for improved generalization under heterogeneous operating conditions.

The absence of validation on real operational turbofan maintenance data remains an important limitation. Although the NASA C-MAPSS benchmark is widely adopted for comparative PHM research, it cannot fully reproduce sensor uncertainty, maintenance interventions, environmental variability, and fleet heterogeneity encountered in real-world aerospace systems.

### 5.6. Future research directions

Several directions can be explored to further improve the proposed prognostic framework. Future studies should validate the model on more complex operating scenarios, particularly the FD002–FD004 subsets and real industrial turbofan datasets, to better assess generalization capability under varying operating conditions and fault mechanisms. In addition, integrating attention mechanisms or Transformer-based temporal architectures may improve long-range degradation modeling performance. Uncertainty-aware approaches such as Bayesian deep learning and ensemble strategies could also be incorporated to provide confidence estimation for RUL predictions and support risk-informed maintenance decisions. Furthermore, transfer learning and domain adaptation techniques may help reduce the gap between simulated benchmark environments and real operational systems, thereby improving practical deployability in aerospace maintenance applications.

### 5.7. Engineering interpretation of the findings

From an engineering perspective, the explainability analyses indicate that the proposed hybrid framework primarily relies on degradation-sensitive sensor variables associated with the High-Pressure Compressor subsystem. Variations in temperature- and pressure-related sensor signals reflect physically meaningful degradation behavior such as efficiency reduction, pressure instability, and progressive compressor wear. The SHAP and LIME results demonstrate that the learned feature representations are consistent with known thermodynamic degradation mechanisms rather than purely statistical correlations. This agreement between model behavior and engineering principles increases confidence in the practical applicability and interpretability of the proposed framework for real-world prognostic and maintenance decision-support systems.

### 5.8. Quantitative analysis in Relation to research

The acquired results were compared with a few recent deep learning-based research published in the turbofan RUL prediction literature in order to further assess the efficacy of the suggested methodology. The suggested hybrid CNN–GRU model performs competitively against a number of cutting-edge methods created using the NASA C-MAPSS benchmark dataset, as shown in Table 15. Because current works frequently differ in preprocessing techniques, RUL labeling methodologies, train–validation protocols, operational subset selection, and assessment settings, it should be highlighted that direct numerical comparisons among studies must be understood carefully. Therefore, the objective of this comparison is not to establish strict ranking superiority, but rather to position the proposed framework within the current literature while emphasizing its robustness-oriented evaluation strategy. The proposed model achieved RMSE = 13.1 ± 0.4, MAE = 9.4 ± 0.3, MAPE = 13.8 ± 0.5%, and $R^2$ = 0.97 ± 0.01 on the FD001 subset averaged over repeated experimental runs. Many of these studies rely on single-run reporting or do not include statistical variability analysis, despite the fact that a number of specialized Transformer-based and advanced temporal convolution architectures claim somewhat lower RMSE values.

As shown in Table 15, recent Transformer-based and advanced temporal convolution architectures generally achieve lower RMSE values due to their stronger capability in modeling long-range temporal dependencies and extracting multi-scale degradation features. In particular, the Transformer-based approach proposed by Zhai et al. demonstrated effective long-sequence representation learning through tendency retainment mechanisms, while the GCSU-ITCN framework introduced by Gan et al. improved temporal convolutional feature extraction using gated channel-selection operations. Nevertheless, many existing studies primarily focus on predictive accuracy and provide limited analysis regarding robustness, repeated-run reproducibility, and explainability. In contrast, the present study emphasizes reliability-oriented evaluation together with SHAP/LIME-based interpretability and statistical consistency analysis.

### 5.9. Uncertainty Awareness and future reliability perspective

Although the proposed Hybrid CNN–GRU framework demonstrated promising prognostic accuracy, robustness, and explainability performance, uncertainty-aware prognostic modeling remains an important future research direction for trustworthy industrial deployment. In practical aero-engine monitoring environments, sensor uncertainty, operational variability, and degradation heterogeneity may introduce prediction uncertainty that cannot be fully captured through deterministic deep learning models alone.

One potential extension involves the integration of Monte Carlo dropout mechanisms during inference to estimate predictive uncertainty distributions. By performing multiple stochastic forward passes through the trained network, confidence intervals associated with RUL predictions can be estimated, thereby providing additional reliability information for maintenance decision-making processes. Such uncertainty-aware prognostic outputs may help distinguish between highly confident predictions and uncertain degradation trajectories under noisy or previously unseen operating conditions.

**Table 15**
Performance comparison on datasets with selected state-of-the-art methods.

| Study | Model | RMSE (FD001) | RMSE (FD002) | RMSE (FD003) | RMSE (FD004) | MAE (FD001) | MAPE (%) (FD001) | $R^2$ (FD001) | Dataset |
|---|---|---|---|---|---|---|---|---|---|
| [43] | Attention-based DCNN | 11.81 | – | – | – | – | – | – | FD001 |
| [14] | Raw-data CNN | 17.0 | – | – | – | – | – | – | FD001 |
| [15] | Deep LSTM (piecewise) | ~12–14 | – | – | – | – | – | – | FD001 |
| [19] | Optimized Random Forest | 19.077 | – | – | – | – | – | 0.789 | FD001 |
| [44] | CAE + Attention-LSTM | 14.44 | – | – | – | 10.49 | – | – | FD001 |
| [45] | Attention-LSTM | 12.33 | – | – | – | – | – | – | FD001 |
| [46] | MLP (feature selection) | 11.80 | – | – | – | – | – | – | FD001 |
| [47] | BiRNN-Autoencoder | 13.92 | – | – | – | – | – | – | FD001 |
| [48] | Transformer with Tendency Retainment | 10.7 | – | – | – | – | – | – | FD001 |
| [49] | GCSU-ITCN | 10.9 | – | – | – | – | – | – | FD001 |
| [50] | Transformer + GCU | 13.4 | – | – | – | – | – | – | FD001 |
| [51] | Transformer (ARR) | 11.36 | – | 11.28 | – | – | – | – | FD001, FD003 |
| [52] | STAR (Hierarchical Transformer) | 11.8 | – | – | – | – | – | – | FD001–FD004 |
| [53] | PSA-GHLSTM Transformer | 11.6 | – | – | – | – | – | – | FD001, FD004 |
| [54] | TTSNet (Transformer–TCN) | 11.02 | 13.25 | 11.06 | 18.26 | – | – | – | FD001–FD004 |
| [55] | AD-TCN (Dilated TCN) | 10.5 | – | – | – | – | – | – | FD001–FD004 |
| [56] | SAE-TCN | 18.01 | – | – | – | – | – | – | FD001 |
| [29] | Dilated LSTM with Attention | 13.8 | – | – | – | – | – | – | FD001 |
| **Proposed (This study)** | **Hybrid CNN–GRU** | **13.1 ± 0.4** | **16.8 ± 0.6** | **14.5 ± 0.5** | **18.9 ± 0.7** | **9.4 ± 0.3** | **8.1 ± 0.3** | **0.97 ± 0.01** | **FD001–FD004** |

Another promising direction involves Bayesian deep learning formulations, such as Bayesian CNN–GRU architectures, where model parameters are represented probabilistically rather than deterministically. Bayesian formulations can improve uncertainty quantification capability while reducing overconfident predictions in safety–critical prognostic applications. In addition, confidence-aware prediction strategies may support risk-sensitive maintenance scheduling and enhance the practical trustworthiness of AI-assisted predictive maintenance systems.

Therefore, future studies may extend the proposed REE framework toward uncertainty-aware and probabilistic prognostic learning paradigms integrating prediction accuracy, robustness stability, explainability consistency, and predictive uncertainty estimation within a unified trustworthy AI framework for industrial prognostics.

## 6. Conclusion

This study used the NASA C-MAPSS benchmark dataset to offer a hybrid CNN–GRU framework for predicting the remaining useful life of turbofan engines. The suggested method captures both long-term operational dependencies and local degradation features by combining the temporal sequence learning capabilities of gated recurrent units with the feature extraction capabilities of convolutional neural networks. The study concentrated on reliability-oriented evaluation using robustness analysis, repeated experimental runs, explainability assessment, and engineering interpretation in addition to predicted accuracy. Extensive tests on the FD001–FD004 subsets showed that, under various operating settings and failure modes, the hybrid CNN–GRU model outperformed solo CNN and GRU designs in terms of prediction performance. The significance of temporal segmentation and regularization techniques in prognostic modeling was further emphasized by window-length and hyperparameter sensitivity analysis. Additionally, SHAP and LIME studies improved model transparency and physical plausibility by offering comprehensible insights about sensor significance and degradation-related temporal behavior. By combining predictive performance with robustness and interpretability considerations, the suggested REE (Reliability–Explainability–Engineering Interpretation) paradigm helps create more reliable deep learning-based prognostic systems. However, the study is constrained by the use of simulated benchmark data; future research will concentrate on uncertainty quantification, transfer learning-based generalization across varied operating settings, and validation using actual industrial aero-engine datasets.

## CRediT authorship contribution statement

**Yıldırım Özüpak:** Writing – review & editing, Writing – original draft, Supervision, Methodology.

## Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## References

[1] M. H. Zubayer, C. Zhang, W. Liu, Y. Wang, and H. M. Imdadul, “Automatic Defect Detection of Jet Engine Turbine and Compressor Blade Surface Coatings Using a Deep Learning-Based Algorithm,” *Coatings 2024, Vol. 14,* vol. 14, no. 4, Apr. 2024, doi: 10.3390/COATINGS14040501.

[2] M. Sobhana, S. Ravi, V. C. Krishna, P. G. V. Koushik, and S. Rajeswari, “Estimation of RUL for Aircraft Engine Using LSTM,” *Proceedings of the 4th International Conference on Smart Electronics and Communication, ICOSEC 2023*, pp. 1701–1707, 2023, doi: 10.1109/ICOSEC58147.2023.10276096.

[3] T. Kebabsa, M. K. Babouri, A. Djebala, and N. Ouelaa, “Advanced diagnostic techniques for turbo compressors: A spectral analysis approach for preventive maintenance,” *Advances in Mechanical Engineering*, vol. 16, no. 5, May 2024, doi: 10.1177/16878132241252329;WEBSITE:WEBSITE:SAGE;JOURNAL:JOURNAL:ADEA;REQUESTEDJOURNAL:JOURNAL:ADEA;ISSUE:ISSUE:DOI.

[4] Y. Özüpak, E. Aslan, and I. Zaitsev, “Explainable LSTM-AdamW based fault diagnosis of aircraft rotating components using airborne acoustic signals under dynamic operating conditions,” *Scientific Reports 2026 16:1*, vol. 16, no. 1, pp. 11449-, Feb. 2026, doi: 10.1038/s41598-026-41889-2.

[5] Y. Liu, Z. Liu, H. Zuo, H. Jiang, P. Li, and X. Li, “A DLSTM-Network-Based Approach for Mechanical Remaining Useful Life Prediction,” *Sensors 2022, Vol. 22,* vol. 22, no. 15, Jul. 2022, doi: 10.3390/S22155680.

[6] Y. Özüpak, E. Aslan, An enhanced interpretable ML approach for forecasting long-term CO2 and N2O emissions via optimized random forest modeling, Fuel 427 (Jan. 2027) 139723, https://doi.org/10.1016/J.FUEL.2026.139723.

[7] Y. Özüpak, F. Alpsalaz, E. Aslan, H. Uzel, High Accuracy Dissolved Oxygen Prediction in Water Quality Analysis with A-XG-Q Hybrid Model, N. Z. J. Mar. Freshwater Res. 60 (1) (Mar. 2026) e70032, https://doi.org/10.1002/NZM2.70032;WEBSITE:WEBSITE:RSNZ;REQUESTEDJOURNAL:JOURNAL:11758805;WGROUP:STRING:PUBLICATION.

[8] K. Chandran, P. K. Shukla, T. A. Ahanger, and R. ul alam Bhat, “Industrial control system design of optimized fopid controller with anomaly detection based on walrus optimization and Bi-LSTM,” *Forschung im Ingenieurwesen 2024 89:1*, vol. 89, no. 1, pp. 1-, Dec. 2024, doi: 10.1007/S10010-024-00765-Z.

[9] M. G. De Giorgi, N. Menga, and A. Ficarella, “Exploring Prognostic and Diagnostic Techniques for Jet Engine Health Monitoring: A Review of Degradation Mechanisms and Advanced Prediction Strategies,” *Energies 2023, Vol. 16,* vol. 16, no. 6, Mar. 2023, doi: 10.3390/EN16062711.

[10] S. Selvaraj, J. Narayanan Thulasi, M. L. J. Johnrose Vijayakumari, and K. Selvaraj, “Deep residual ensemble model for predicting remaining useful life of turbo fan engines,” *International Journal of Turbo and Jet Engines*, vol. 42, no. 2, pp. 229–239, May 2025, doi: 10.1515/TJJ-2024-0025/XML.

[11] N. Menga, A. Mothakani, M. G. De Giorgi, R. Przysowa, and A. Ficarella, “Extreme Learning Machine-Based Diagnostics for Component Degradation in a Microturbine,” *Energies 2022, Vol. 15,* vol. 15, no. 19, Oct. 2022, doi: 10.3390/EN15197304.

[12] E. Aslan and Y. Özüpak, “A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images,” *Magnetic Resonance Letters*, p. 200275, Apr. 2026, doi: 10.1016/J.MRL.2026.200275.

[13] J. Judeson Antony Kovilpillai, S. S. Mohamed, Pragya, M. H. Mir, T. Singh, and U. K. Singh, "Deep Learning Models for Enhanced RUL Prediction in Turbofan Jet Engines," *Lecture Notes in Networks and Systems*, vol. 1073, pp. 685–699, 2025, doi: 10.1007/978-981-97-5703-9_56.
[14] M.I. Hossain, S. Bin Helal, K. Ray, A. Ara, J.U. Asia, A. Shufian, Remaining Useful Life Prediction for Turbojet Engines using Deep CNN on Raw Sensor Data, in: *2024 27th International Conference on Computer and Information Technology*, 2024, pp. 2098–2103, https://doi.org/10.1109/ICCIT64611.2024.11022015.
[15] O. Asif, S.A.L.I. Haider, S.R. Naqvi, J.F.W. Zaki, K.S. Kwak, S.M.R. Islam, A Deep Learning Model for remaining Useful Life Prediction of Aircraft Turbofan Engine on C-MAPSS Dataset, IEEE Access 10 (2022) 95425–95440, https://doi.org/10.1109/ACCESS.2022.3203406.
[16] S. Sharma, A. K. Pandit, and S. Sharanya, "Predicting Aircraft Turbofan Engine Degradation with Recurrent Neural Networks," *2024 IEEE International Conference on Information Technology, Electronics and Intelligent Communication Systems, ICITEICS 2024*, 2024, doi: 10.1109/ICITEICS61368.2024.10625502.
[17] U. Thakkar and H. Chaoui, "Remaining Useful Life Prediction of an Aircraft Turbofan Engine Using Deep Layer Recurrent Neural Networks," *Actuators 2022, Vol. 11,* vol. 11, no. 3, Feb. 2022, doi: 10.3390/ACT11030067.
[18] A. Peringal, M. B. Mohiuddin, and A. Hassan, "Remaining Useful Life Prediction for Aircraft Engines using LSTM," Jan. 2024, Accessed: Jan. 31, 2026. [Online]. Available: https://arxiv.org/pdf/2401.07590.
[19] R.K. Gupta, J. Nakum, P. Gupta, A Machine Learning Approach for Turbofan Jet Engine Predictive Maintenance, Procedia Comput. Sci. 259 (Jan. 2025) 161–171, https://doi.org/10.1016/J.PROCS.2025.03.317.
[20] L. R. de Freitas Júnior and F. G. Guimarães, "Remaining Useful Life Prediction in an Aerospace Engine: A Multivariable Fuzzy Time Series Classification Approach," *Expert Syst.*, vol. 43, no. 2, p. e70202, Feb. 2026, doi: 10.1111/EXSY.70202;WEBSITE:WEBSITE:PERICLES;REQUESTEDJOURNAL:JOURNAL:14680394;ISSUE:ISSUE:DOI.
[21] L. Shen, Y. Wang, B. Du, H. Yang, and H. Fan, "Remaining Useful Life Prediction of Aero-Engine Based on Improved GWO and 1DCNN," *Machines 2025, Vol. 13,* vol. 13, no. 7, Jul. 2025, doi: 10.3390/MACHINES13070583.
[22] M. Wu, H. Zhou, R. Yao, and S. Ren, "Life prediction and health assessment of aero-engine gas path using digital twin and deep learning," *Complex & Intelligent Systems 2025 11:9*, vol. 11, no. 9, pp. 399-, Jul. 2025, doi: 10.1007/S40747-025-02027-Z.
[23] S. Szrama, "Turbofan engine health status prediction with heterogeneous ensemble deep neural networks," *International Journal of Data Science and Analytics 2025 22:1*, vol. 22, no. 1, pp. 23-, Jan. 2026, doi: 10.1007/S41060-025-00989-4.
[24] S. Szrama, G. Szymański, D. Mokrzan, "Aircraft propulsion health status prognostics and prediction," *advances in Science and Technology*, Research Journal 19 (5) (2025) 321–335, https://doi.org/10.12913/22998624/202232.
[25] S. Szrama, Adaptive cluster-count selection via deep Q-learning for turbofan engine prognostics and health monitoring, Neurocomputing 665 (Feb. 2026) 132294, https://doi.org/10.1016/J.NEUCOM.2025.132294.
[26] S. Szrama, T. Lodygowski, S. Szrama, T. Lodygowski, Turbofan engine health status prediction with artificial neural network, Aviation 28 (4) (Dec. 2024) 225–234, https://doi.org/10.3846/AVIATION.2024.22554.
[27] T. Lodygowski and S. Szrama, "Unsupervised Classification and Remaining Useful Life Prediction for Turbofan Engines Using Autoencoders and Gaussian Mixture Models: A Comprehensive Framework for Predictive Maintenance," *Applied Sciences 2025, Vol. 15,* vol. 15, no. 14, Jul. 2025, doi: 10.3390/APP15147884.
[28] S. Szrama, Optimizing aircraft engine longevity: a comparative framework for dynamically adaptive predictive maintenance using autoencoders, LSTMs, and Gaussian processes, Eng. Appl. Artif. Intell. 156 (Sep. 2025) 111199, https://doi.org/10.1016/J.ENGAPPAI.2025.111199.
[29] A. Boujamza, S.L. Elhaq, Remaining useful life predictions of aero-engines using dilated LSTM with attention mechanism, Franklin Open 14 (Mar. 2026) 100514, https://doi.org/10.1016/J.FRAOPE.2026.100514.
[30] A. Ayodeji, et al., Causal augmented ConvNet: a temporal memory dilated convolution model for long-sequence time series prediction, ISA Trans. 123 (Apr. 2022) 200–217, https://doi.org/10.1016/J.ISATRA.2021.05.026.
[31] J. Shang, D. Xu, P. Liang, C. Jiang, H. Qiu, L. Gao, Out-of-domain generalization for remaining useful life prediction of rotating machinery from a single source: an adversarial contrastive learning approach, Mech. Syst. Signal Process. 236 (Aug. 2025) 112965, https://doi.org/10.1016/J.YMSSP.2025.112965.
[32] J. Shang, D. Xu, H. Qiu, C. Jiang, L. Gao, E. Zio, CEC-FedISDG: a Cloud-Edge Collaboration Federated Invariance and Specificity Domain Generalization Method for machine remaining Useful Life Prediction, IEEE Trans. Syst. Man Cybern. Syst. (2026), https://doi.org/10.1109/TSMC.2026.3663567.
[33] J. Shang, D. Xu, H. Qiu, L. Gao, C. Jiang, P. Yi, A novel data augmentation framework for remaining useful life estimation with dense convolutional regression network, J. Manuf. Syst. 74 (Jun. 2024) 30–40, https://doi.org/10.1016/J.JMSY.2024.02.011.
[34] "NASA Turbofan Jet Engine Data Set." Accessed: Jan. 31, 2026. [Online]. Available: https://www.kaggle.com/datasets/behrad3d/nasa-cmaps.
[35] Y. Özüpak, S. Mansurov, Optimizing electricity demand forecasting with a novel RNN-LSTM hybrid model, Energy Sources Part B 20 (1) (2025) Dec, https://doi.org/10.1080/15567249.2025.2531448.
[36] Y. Özüpak, S. Mansurov, A Hybrid Deep Learning Approach for Multi-output Short-Term Electricity demand forecasting, Concurr. Comput. 37 (25–26) (Nov. 2025) e70356, https://doi.org/10.1002/CPE.70356;SUBPAGE:STRING:FULL.
[37] S. Benkaihoul, S. Khadar, Y. Özüpak, E. Aslan, M. M. Almalki, and M. A. Mossa, "Advanced Fault Classification in Induction Motors for Electric Vehicles Using A Stacking Ensemble Learning Approach," *World Electric Vehicle Journal 2025, Vol. 16,* vol. 16, no. 11, Nov. 2025, doi: 10.3390/WEVJ16110614.
[38] Y. Özüpak, Analysis of the Model Designed for magnetic Resonance based Wireless Power transfer using FEM, Journal of Engineering Research 11 (3) (Sep. 2023) 100–111, https://doi.org/10.36909/JER.17631.
[39] Y. Özüpak, Designing a Reactor for use in High Voltage Power Systems and performing Experimental and simulation Analysis, Journal of Engineering Research 11 (3) (Sep. 2023) 169–182, https://doi.org/10.36909/JER.17017.
[40] Y. Özüpak, Design and analysis of different transformer models used for charging electric vehicles with wireless power transmission system, J. Fac. Eng. Archit. Gazi Univ. 39 (2) (2024) 1113–1122, https://doi.org/10.17341/gazimmfd.1202033.
[41] W. Anping, et al., Bayesian-Driven Optimization of MDCNN-LSTM-RSA: a New Model for predicting Aeroengine RUL, IEEE Trans. Reliab. 74 (4) (2025) 5323–5334, https://doi.org/10.1109/TR.2025.3574975.
[42] Y. özüpak, "Real-time detection of photovoltaic module faults using a hybrid machine learning model," *Solar Energy*, vol. 302, p. 114014, Dec. 2025, doi: 10.1016/J.SOLENER.2025.114014.
[43] A. Muneer, S. M. Taib, S. M. Fati, and H. Alhussian, "Deep-Learning Based Prognosis Approach for Remaining Useful Life Prediction of Turbofan Engine," *Symmetry 2021, Vol. 13,* vol. 13, no. 10, Oct. 2021, doi: 10.3390/SYM13101861.
[44] S. M. Elsherif, B. Hafiz, M. A. Makhlouf, and O. Farouk, "A deep learning-based prognostic approach for predicting turbofan engine degradation and remaining useful life," *Scientific Reports 2025 15:1*, vol. 15, no. 1, pp. 26251-, Jul. 2025, doi: 10.1038/s41598-025-09155-z.
[45] M. Dida, A. Cheriet, and M. Belhadj, "Remaining Useful Life Prediction Using Attention-LSTM Neural Network of Aircraft Engines," *Int. J. Progn. Health Manag.*, vol. 16, no. 2, Jul. 2025, doi: 10.36001/IJPHM.2025.V16I2.4274.
[46] Y. Alomari, M. Andó, and M. L. Baptista, "Advancing aircraft engine RUL predictions: an interpretable integrated approach of feature engineering and aggregated feature importance," *Scientific Reports 2023 13:1*, vol. 13, no. 1, pp. 13466-, Aug. 2023, doi: 10.1038/s41598-023-40315-1.
[47] W. Yu, I.Y. Kim, C. Mechefske, Remaining useful life estimation using a bidirectional recurrent neural network based autoencoder scheme, Mech. Syst. Signal Process. 129 (Aug. 2019) 764–780, https://doi.org/10.1016/j.ymssp.2019.05.005.
[48] Z. Zhai, J. Wen, F. Wang, Z. Zhao, Y. Guo, X. Chen, Remaining Useful Life Prediction of Aero-Engine based on Transformer with Tendency Retainment, in: In *2022 International Conference on Sensing, Measurement and Data Analytics in the Era of Artificial Intelligence, ICSMD 2022 - Proceedings*, Institute of Electrical and Electronics Engineers Inc, 2022, https://doi.org/10.1109/ICSMD57530.2022.10058242.
[49] F. Gan, Y. Qin, B. Xia, D. Mi, L. Zhang, Remaining useful life prediction of aero-engine via temporal convolutional network with gated convolution and channel selection unit, Appl. Soft Comput. 167 (Dec. 2024), https://doi.org/10.1016/j.asoc.2024.112325.
[50] Y. Mo, Q. Wu, X. Li, and B. Huang, "Remaining useful life estimation via transformer encoder enhanced by a gated convolutional unit," *Journal of Intelligent Manufacturing 2021 32:7*, vol. 32, no. 7, pp. 1997–2006, Mar. 2021, doi: 10.1007/S10845-021-01750-X.

[51] G. Kim, J.G. Choi, S. Lim, Using transformer and a reweighting technique to develop a remaining useful life estimation method for turbofan engines, Eng. Appl. Artif. Intell. 133 (Jul. 2024) 108475, https://doi.org/10.1016/J.ENGAPPAI.2024.108475.
[52] Z. Fan, W. Li, and K. C. Chang, "A Two-Stage Attention-Based Hierarchical Transformer for Turbofan Engine Remaining Useful Life Prediction," *Sensors 2024, Vol. 24,* vol. 24, no. 3, Jan. 2024, doi: 10.3390/S24030824.
[53] X. Chen, "A novel transformer-based DL model enhanced by position-sensitive attention and gated hierarchical LSTM for aero-engine RUL prediction," *Scientific Reports 2024 14:1*, vol. 14, no. 1, pp. 10061-, May 2024, doi: 10.1038/s41598-024-59095-3.
[54] Z. Li, S. Luo, H. Liu, C. Tang, and J. Miao, "TTSNet: Transformer–Temporal Convolutional Network–Self-Attention with Feature Fusion for Prediction of Remaining Useful Life of Aircraft Engines," *Sensors 2025, Vol. 25,* vol. 25, no. 2, Jan. 2025, doi: 10.3390/S25020432.
[55] Z. Lu *et al.*, "A multi-scale cross-dimension interaction approach with adaptive dilated TCN for RUL prediction," *Scientific Reports 2025 15:1*, vol. 15, no. 1, pp. 19229-, Jun. 2025, doi: 10.1038/s41598-025-00572-8.
[56] X. Liu, L. Xiong, Y. Zhang, and C. Luo, "Remaining Useful Life Prediction for Turbofan Engine Using SAE-TCN Model," *Aerospace 2023, Vol. 10,* vol. 10, no. 8, Aug. 2023, doi: 10.3390/AEROSPACE10080715.